

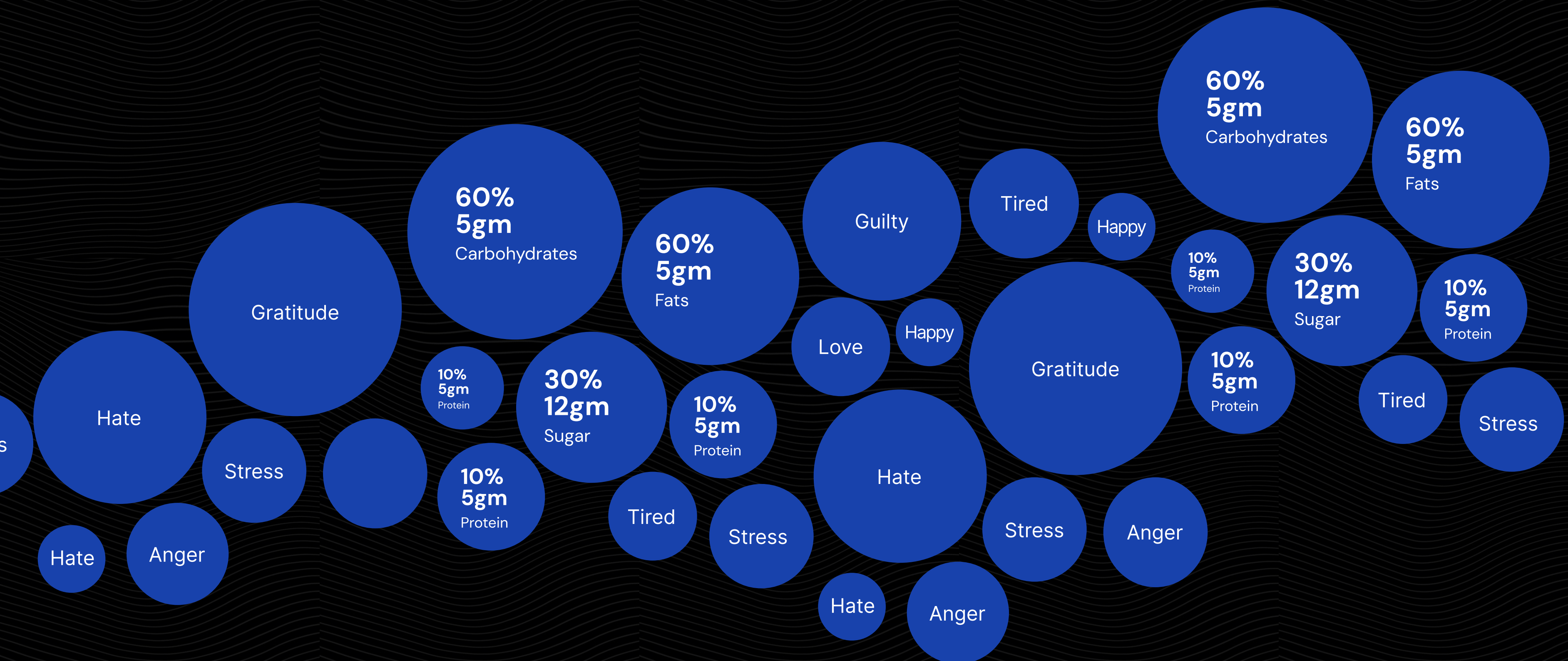
Mãe

Virtual mom: who takes care of your food and emotions



#1 Problem Area

Balancing work, health and your routine is all tied to your what you eat, and we eat our emotions. Along with eating what we like and what it does to your body is also important and the balance is a fine line, between satisfying emotions and nutritional quota.



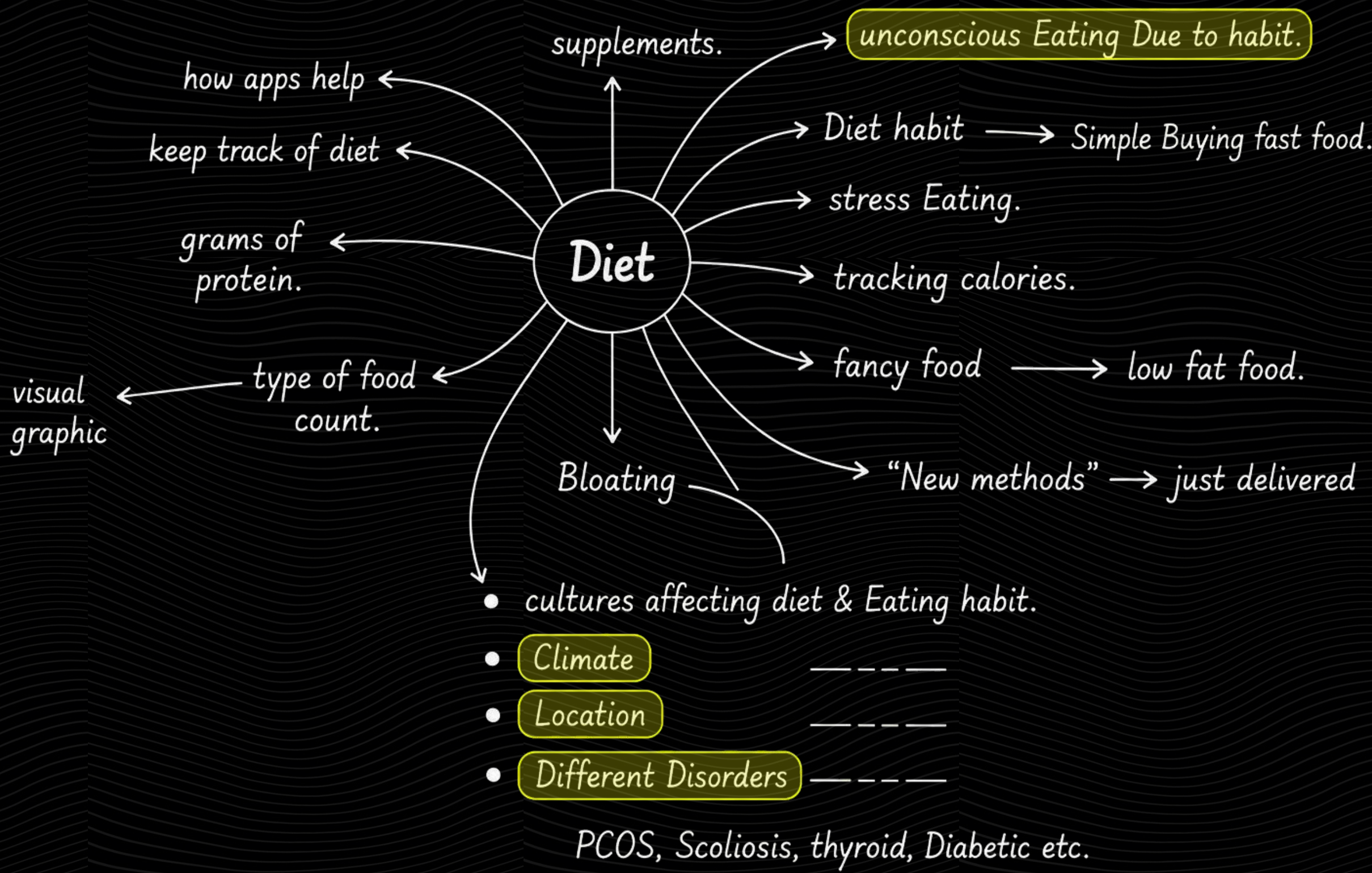
Bubbles of emotions and nutritional information clouded and messed up in a cloud

#2 How Might We?

How might we understand and improve a persons health in a holistic way (ie. body, mind, lifestyle and social life) without overwhelming them with data.

#3 Research Direction

Before we starting with any assumptions on the notion of food and diet, from desk research the following question I tried to understand in the ecosystem of food habits and diets.



★ And understand the unconscious and conscious food choices, cravings, social norms, associations with emotions

Data Collection and Desk Research

Survey

To captured broad trends and common patterns in eating behavior across a diverse group of participants.

Interview

Added depth by understanding personal experiences, motivations, & perceptions behind the patterns observed.

Experiment

Explored social influence by observing discussion opinion shifts, and group dynamics.

Focused Group

Observed real—life behavior to check whether stated preferences translated into actual

The length of container indicates a deductive approach to user understanding, allowing for a deeper analysis of behavioral patterns.

Form Responses

1	Timestamp	Disclaimer	Age	Gender	Place(city) where you grew up	What is your current living situation?	What is your set-up	What is your diet preference?	How many meals do you usually eat
2	1/14/2026 9:46:11	By continuing this survey	40	Male	Faridabad Haryana	With Family		Eggetarian	3 meals
3	1/14/2026 9:52:56	By continuing this survey	23	Female	Patna, Raigarh, Siliguri	Without Family	Paying Guest (PG)	Non-Vegetarian	3 meals + snacks
4	1/14/2026 9:55:19	By continuing this survey	23	Female	Chennai, Tamil Nadu	Without Family	Hostel	Vegetarian	3 meals
5	1/14/2026 9:56:58	By continuing this survey	22	Male	Family	Without Family	Paying Guest (PG)	Vegetarian	1-2 meals + snacks
6	1/14/2026 10:01:36	By continuing this survey	23	Male	Pune	With Family		Non-Vegetarian	3 meals + snacks
7	1/14/2026 10:03:52	By continuing this survey	22	Male	Hyderabad	With Family		Both veg and non veg	3 meals + snacks
8	1/14/2026 10:04:37	By continuing this survey	24	Female	KERALA	Without Family	Paying Guest (PG)	Non-Vegetarian	3 meals + snacks
9	1/14/2026 10:04:58	By continuing this survey	26	Male	Faridabad	With Family		Non-Vegetarian	3 meals
10	1/14/2026 10:05:13	By continuing this survey	30	Female	Vizag	Without Family	Paying Guest (PG)	Non-Vegetarian	Mostly snacks, no fixed meals
11	1/14/2026 10:05:49	By continuing this survey	24	Male	Karad	Without Family	Paying Guest (PG)	Vegetarian	1-2 meals + snacks
12	1/14/2026 10:07:29	By continuing this survey	25	Male	Hyderabad	Without Family	Hostel	Non-Vegetarian	3 meals + snacks

113 responses

Summary

Question

Individual

Disclaimer

113 responses

Copy chart

By continuing this survey, you agree that your responses will be used only for academic research.

113 (100%)

Profile	Emotional Eating	Environment-Driven	Habitual	Reel Impact
Foody	1		1	High → Strong craving + frequent action (ordering / eating)
Diet Conscious	0		0	Medium → Desire is triggered but behaviour is cognitively restrained
Health Issue	1		1	Low
Moody	0		0	Low → Emotional state dominates eating, reels have weak influence
Neutral	0		1	Medium → Situational curiosity, occasional behavioural response
Food reels only work when habitual control is weak.				
Gender	Emotional Eating	Environment-Driven	Habit / Routine	
Female (6)	3	2		1
Male (4)	1	1		2
Females show emotion-linked eating				
Males show habitual & functional eating				
Exposure Type	Hunger Increased	No Change	Hunger Dissipated	
Food Reels	2	3		0
Neutral Content	0	4		1
Only food reels caused hunger increase → confirms media effect.				
Variable	Category	Frequency		
Total Participants				10
Gender	Female			6
Gender	Male			4
Exposure Type	Food Reels			5
Exposure Type	Neutral Content			5
Choice	Group A (Food Exposure)	Group B (Neutral Exposure)		
	Shreye, Annyesha, Atharva	Jai, Palak, Sharanya		
Chips				
Peanuts	Pritik, Amey	Kanishka, Vaibhavi		



1. Survey responses
2. Experiment data set: identified eating persona

For detailed analysis refer to the following links:

1. [Data analysis report](#)
2. [Experiment analysis](#)
3. [Interview analysis](#)

#4 Data Insights

113

Survey Responses

19

Interviews

Focused Group

2

10

Experimental Participants

Women

Conscious Awareness

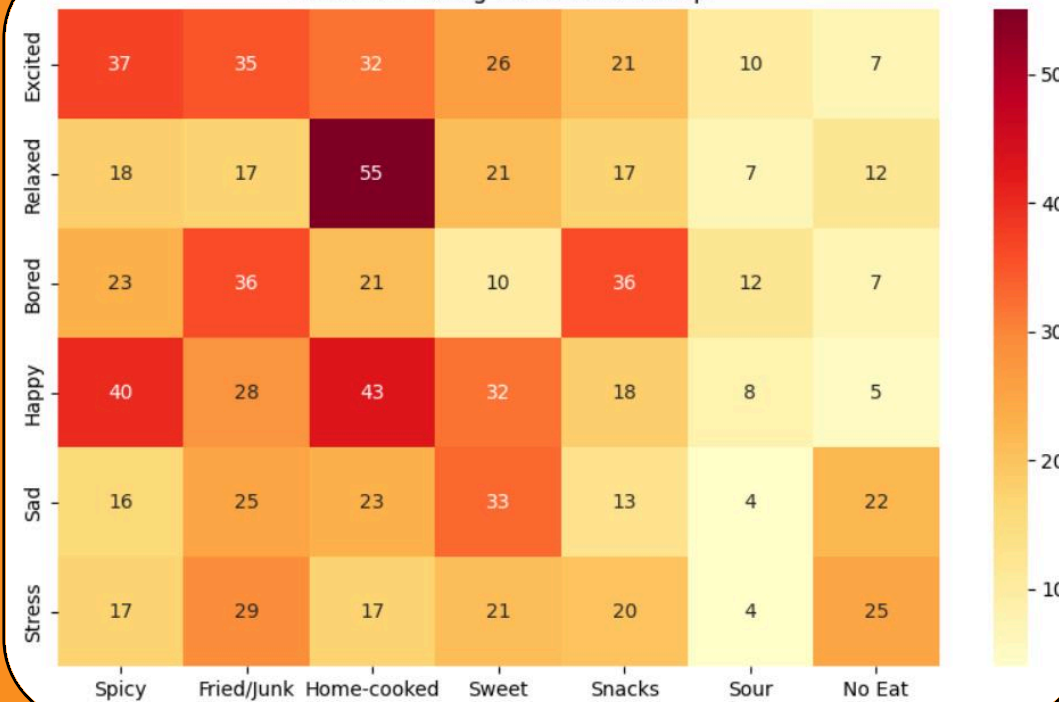
Unconscious Tendencies

Men

4 Personas

- Stress over-eaters
- Stress under-eaters
- Balanced copers
- Sensory Seekers

Emotional Eating Patterns Heatmap



Chi-square value: 123.81182799916223
Degrees of freedom: 30
P-value: 2.329575855894264e-13

Reject H0: Food choice depends on emotional state

The statistically significant chi-square results and a p-value well below 0.05 strongly suggest that food choices are driven by emotional states. This confirms that the relationship is robust and highly unlikely to have occurred by random chance.

52

Stress Overeaters

9

Balanced Copers

7

Sensory Seekers

25

Stress Under eaters

Population of Personas

12

Daily Structure & Routine-Driven Eating

13

Home vs Institutional / Outside Food Negotiation

Thematic Cloud

Emotional Regulation Through Food

Digital & Visual Media Influence on Food Perception

14

13



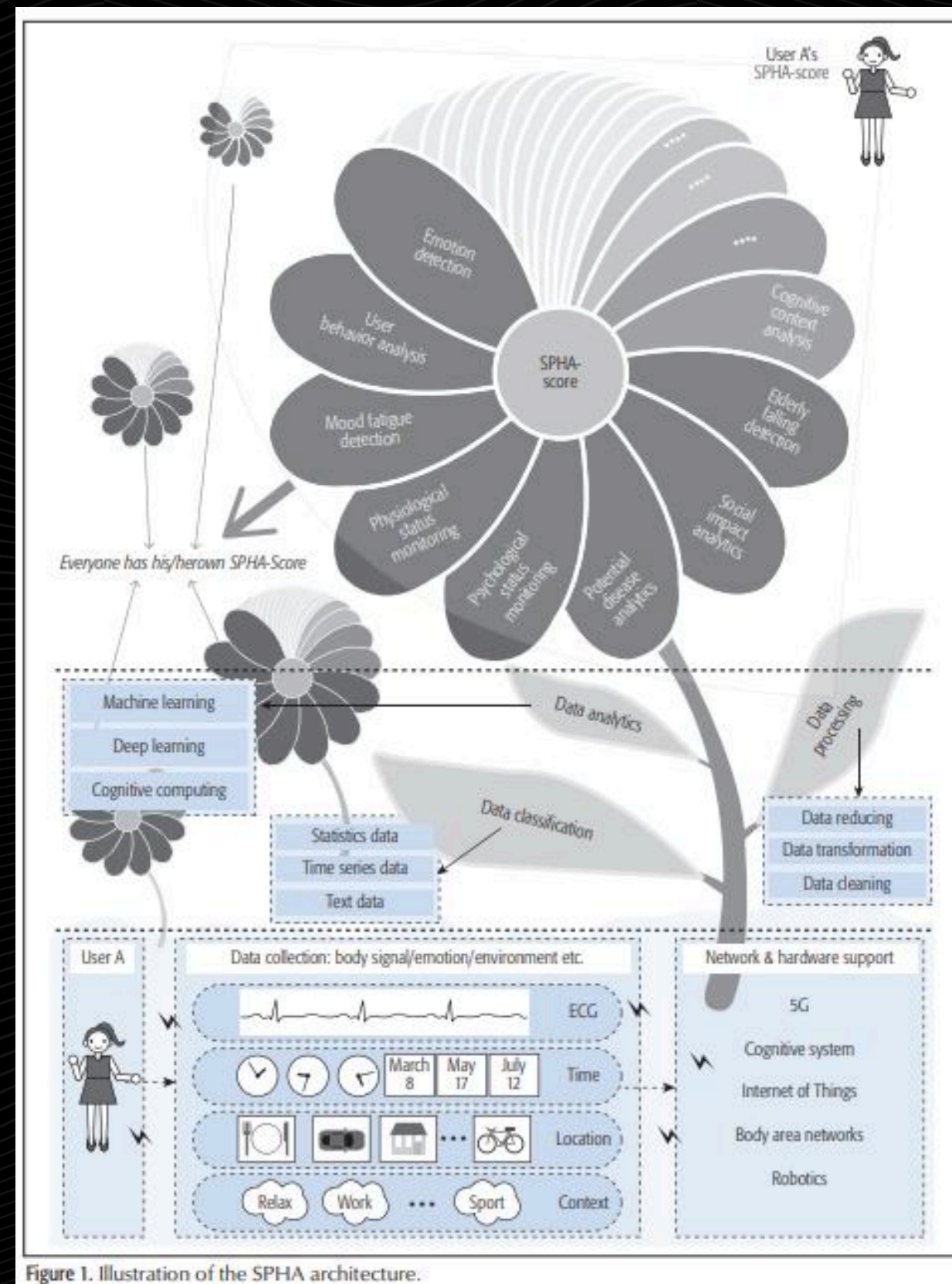
Data analysis insights:

1. Strong emotional attachment to food
2. Different personas and their different behaviors and response to emotional trigger
3. Male vs female unconscious tendencies

#5 Design Direction & App Concept

An app that can monitor people's emotional wellbeing and integrate it with food behaviors and eating pattern and guide them in direction to their health goals.

App concept



SPHA (Smart Personal Health Advisor)

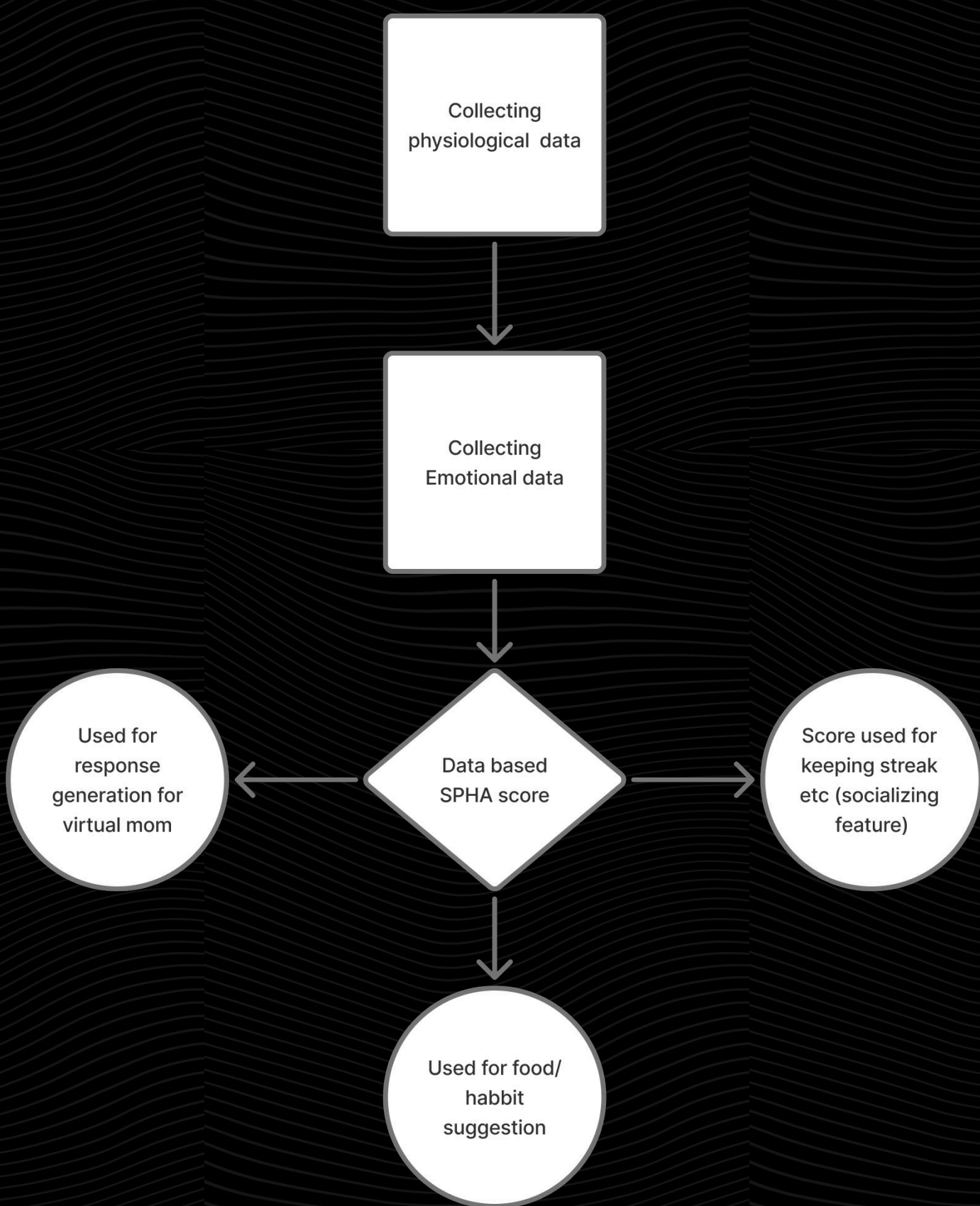
This paper shows how to calculate the emotional wellbeing and calculate a score as SPHA score



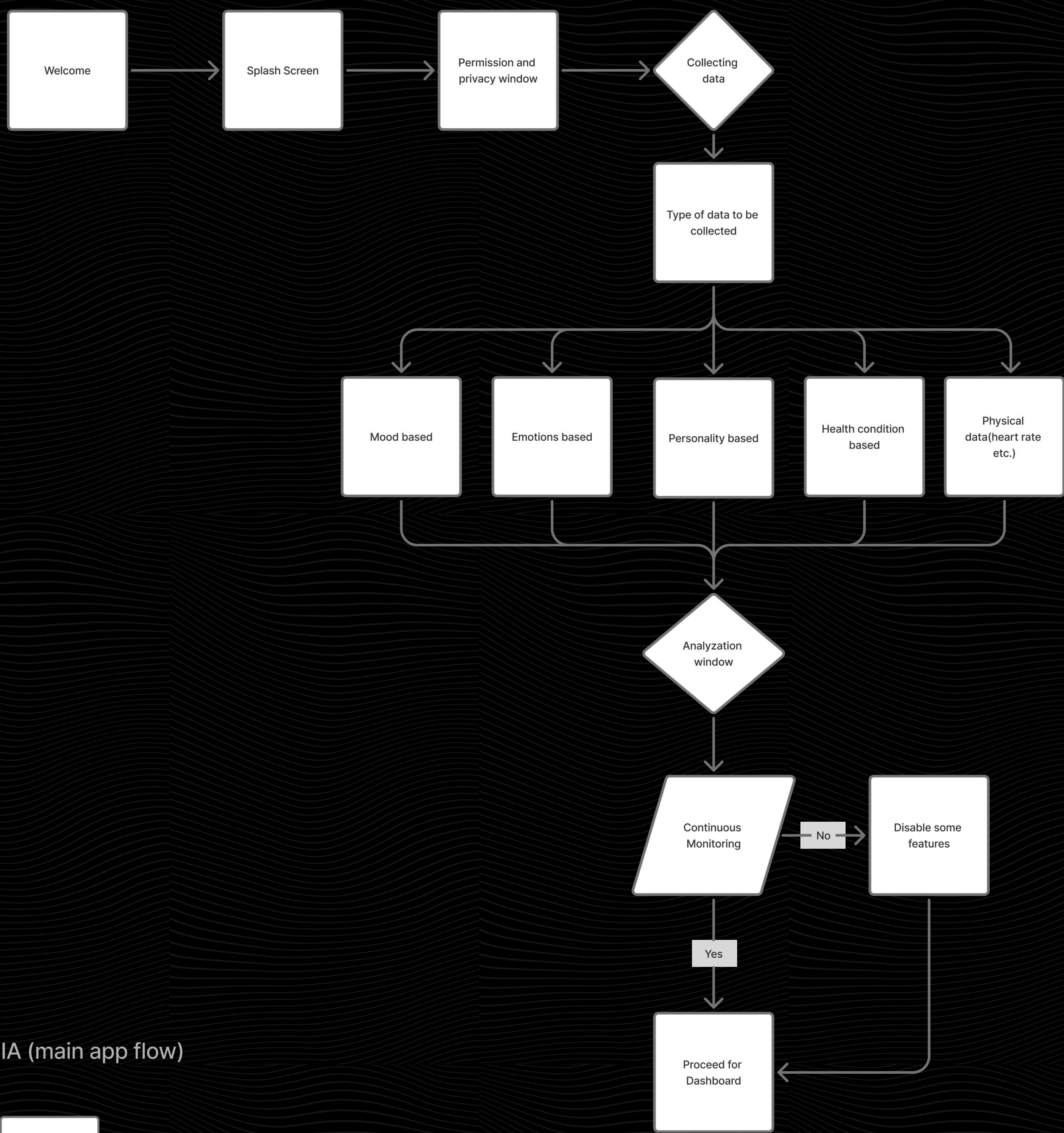
The app takes inspiration from SPHA to look after their emotional wellbeing and machine learning agent to map eating behaviors, cravings and patterns.

#6 Information Architecture

Information architecture for data collection and utilization



Information Architecture for initializing the app



IA (main app flow)

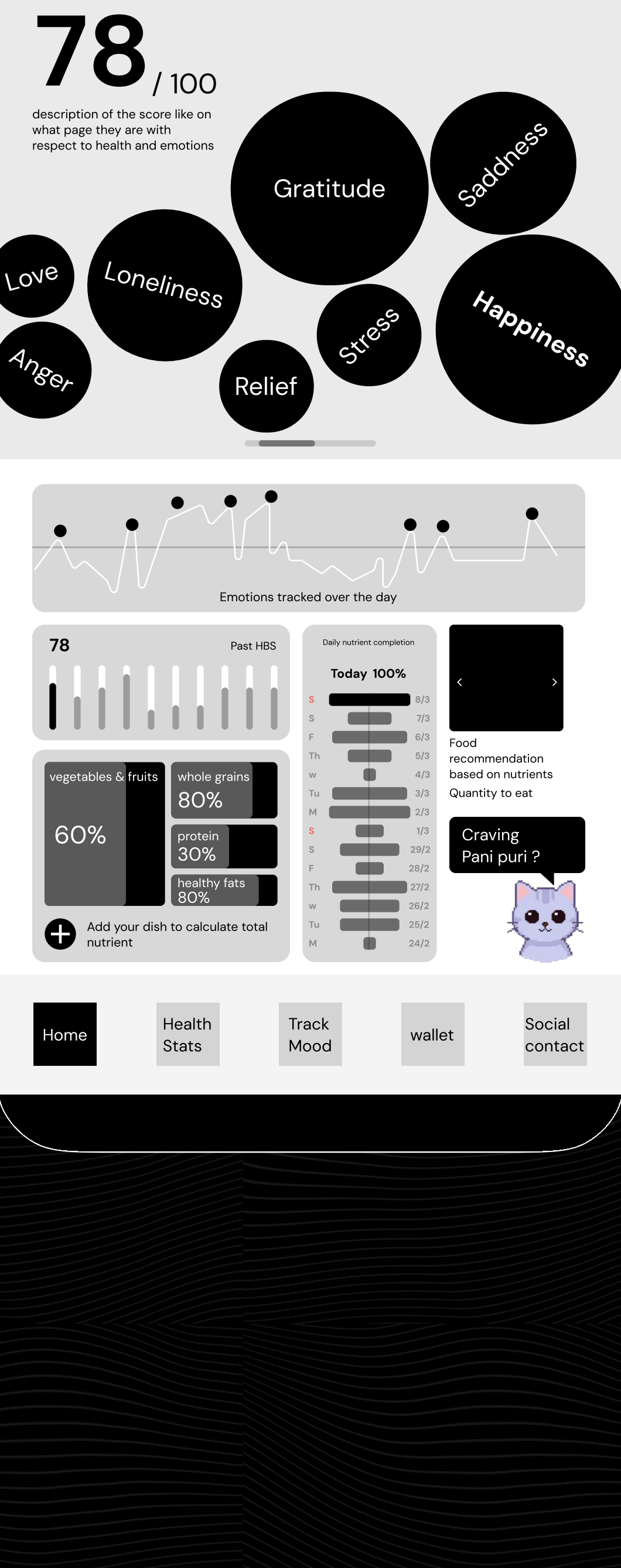


#7 Features & Wireframes

The features app provides to function as a one complete buddy or “virtual mom” are below each one is important for the concept of SPHA to work effectively and turn body vital eating patters data into a personalized food guide

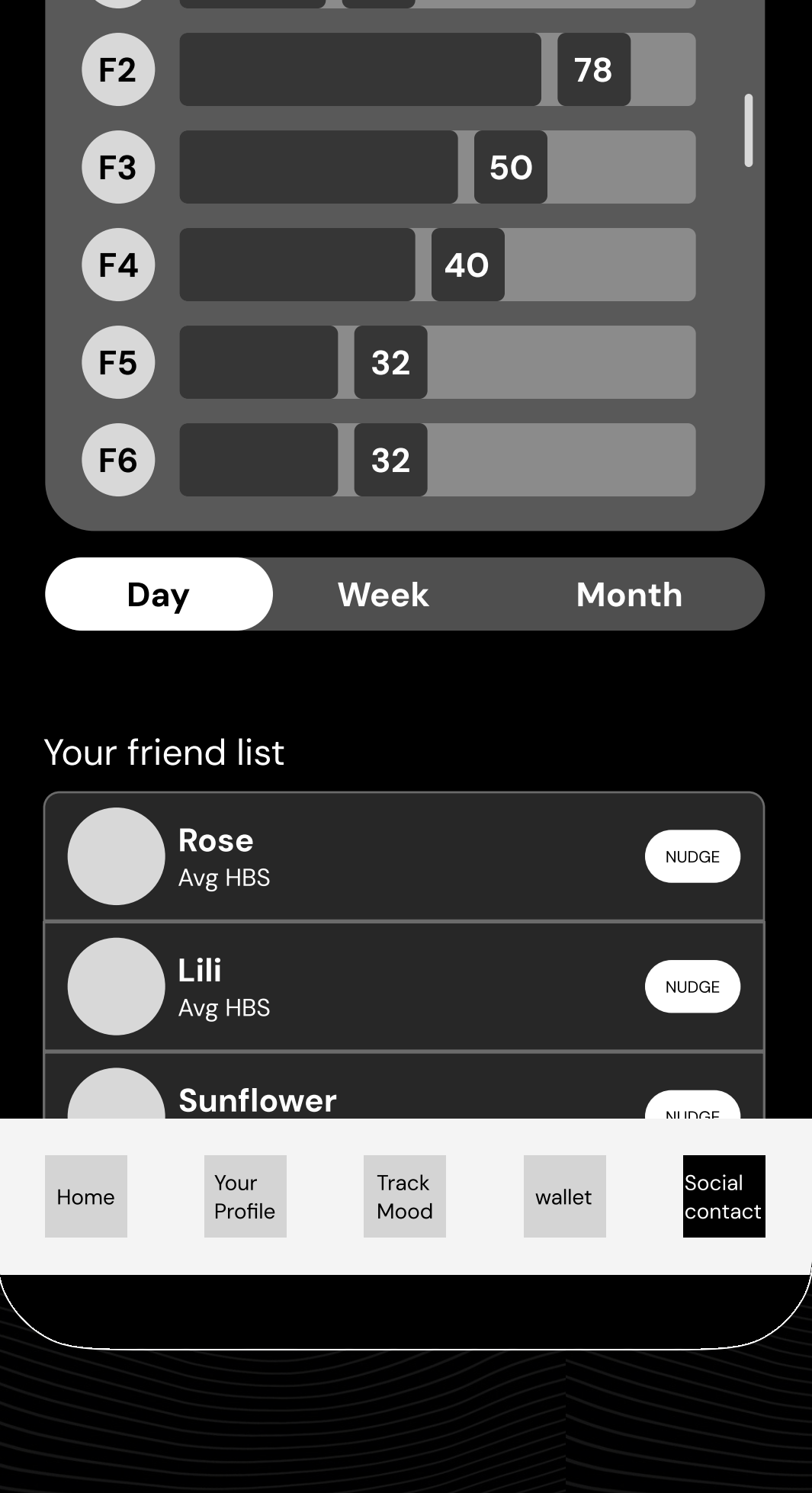
01 Health Balance Score

Description of the score like on what page they are with respect to health and emotions



SPHA score as “Health Balance Score”

The data is collected through smart watch or series of small light questions asked through out the day

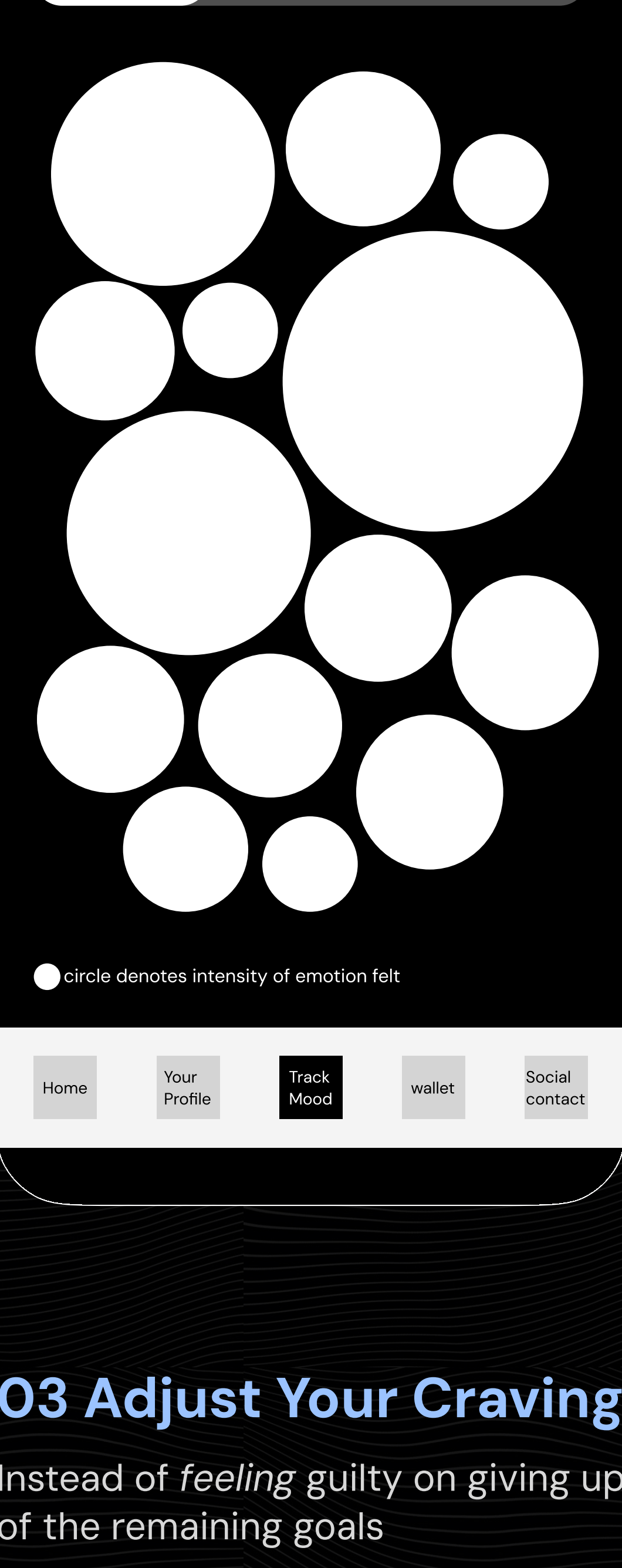


You could maintain streak with your friends and nudge them to maintain the same

Leader board

Nudge/ Reminder to your friend

02 Emotion Popping As Bubbles

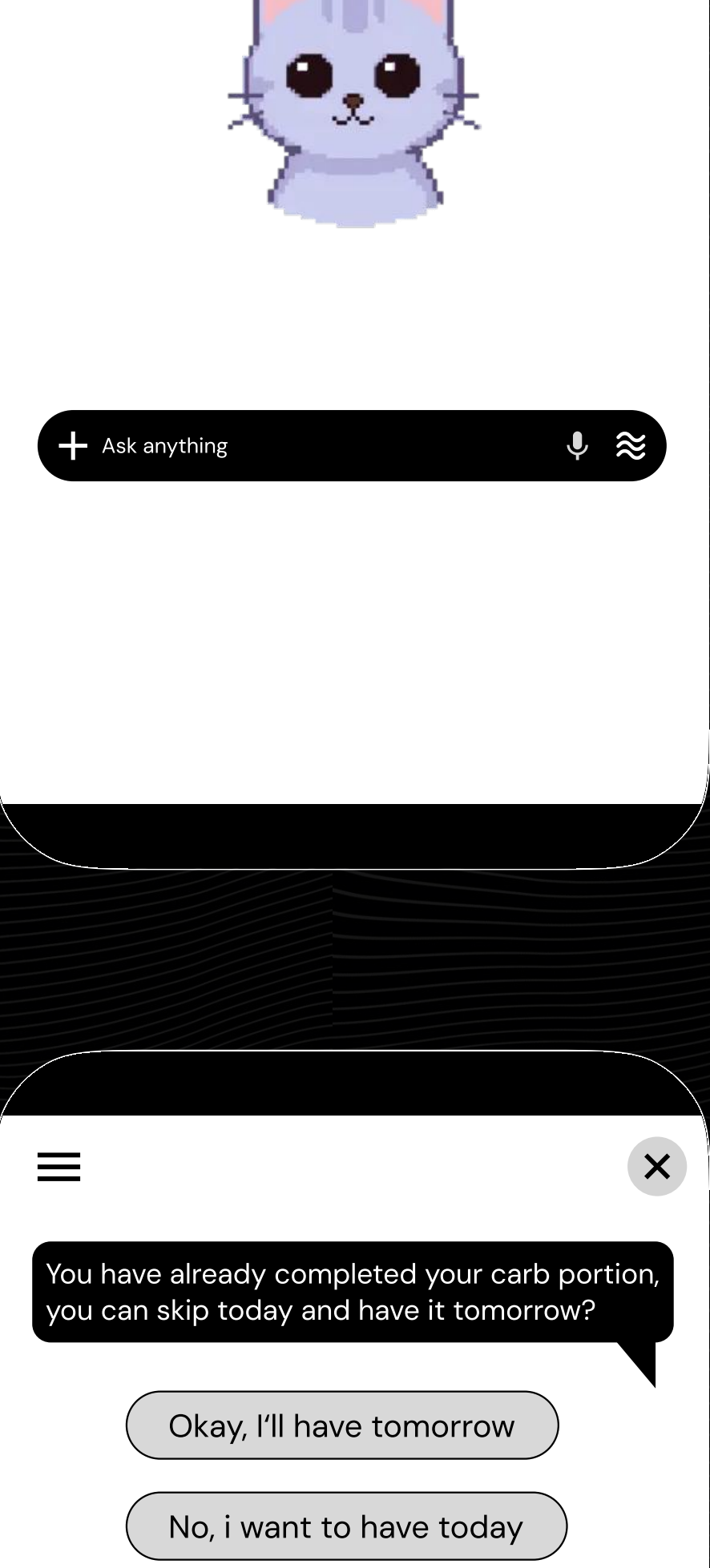


Data Insight Conversion

Emotions as bubbles to reflect the strong correlation of food and emotion.

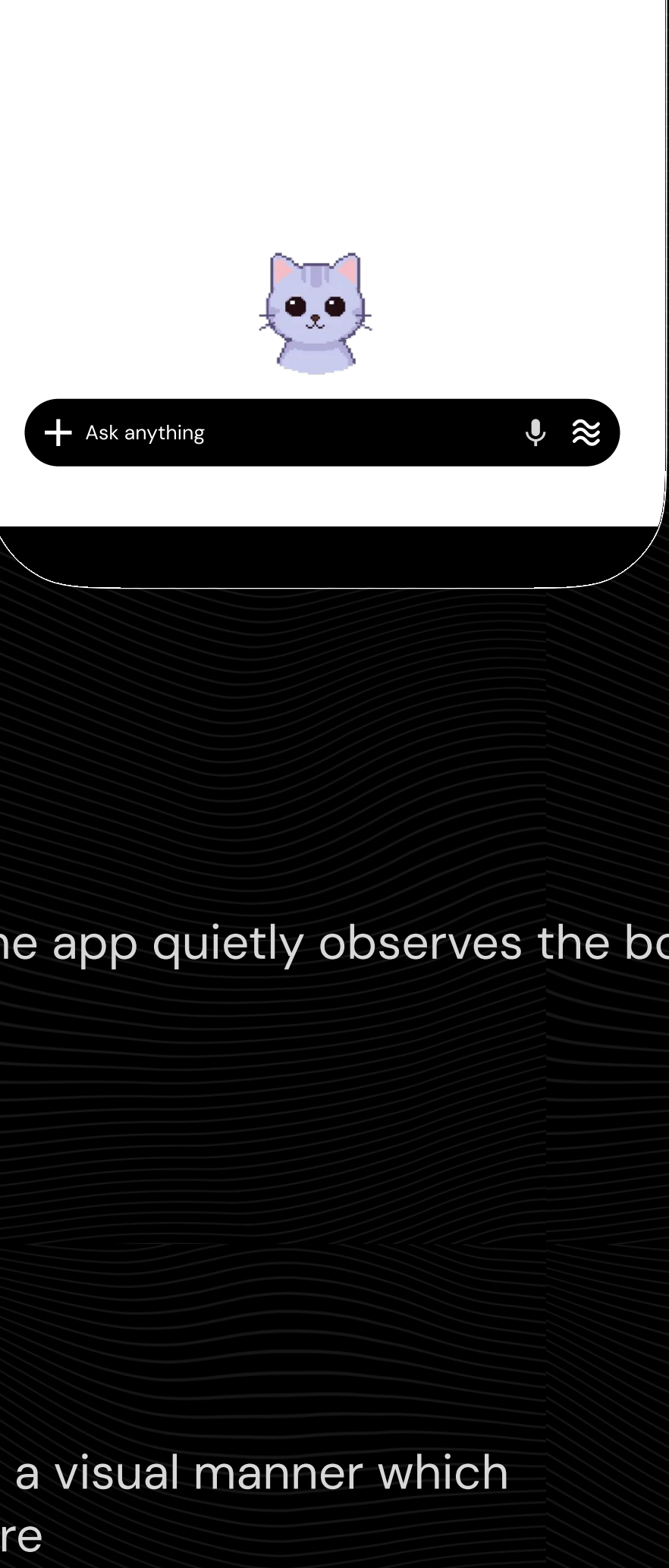
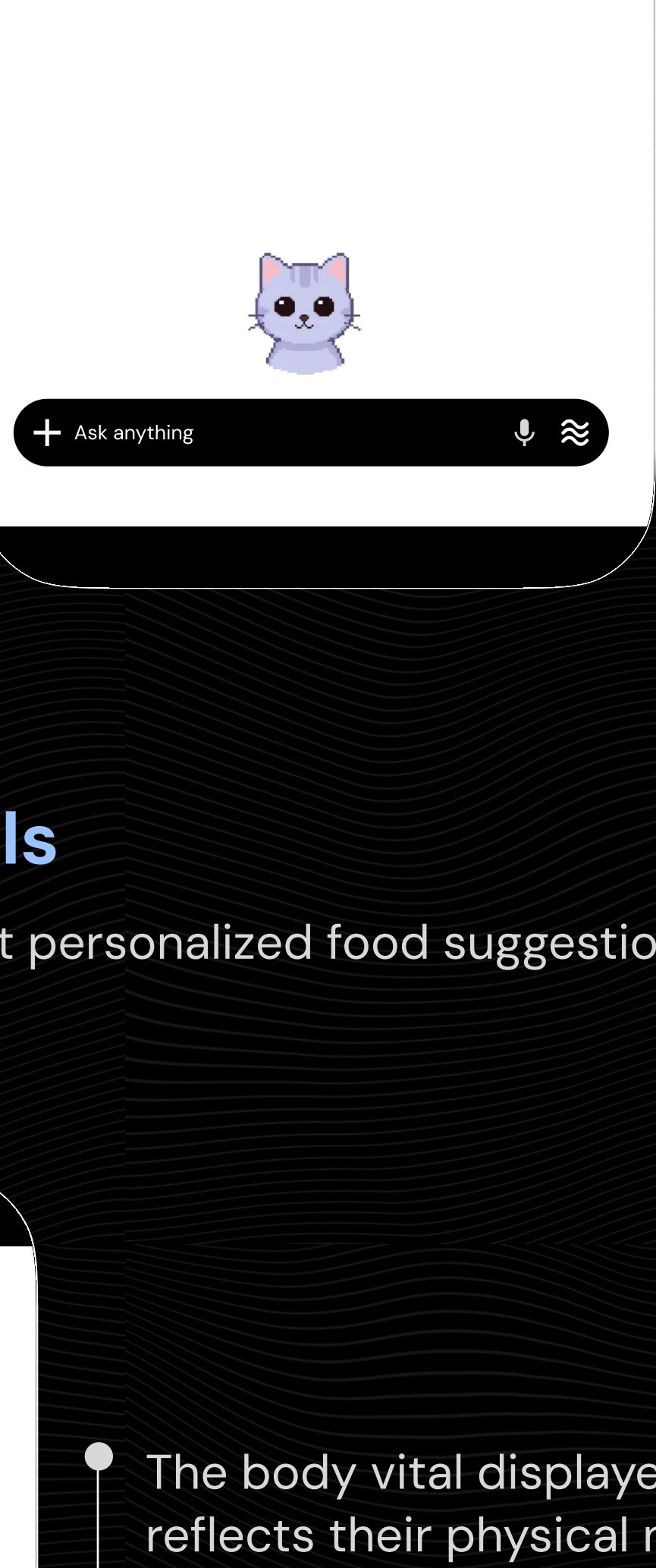
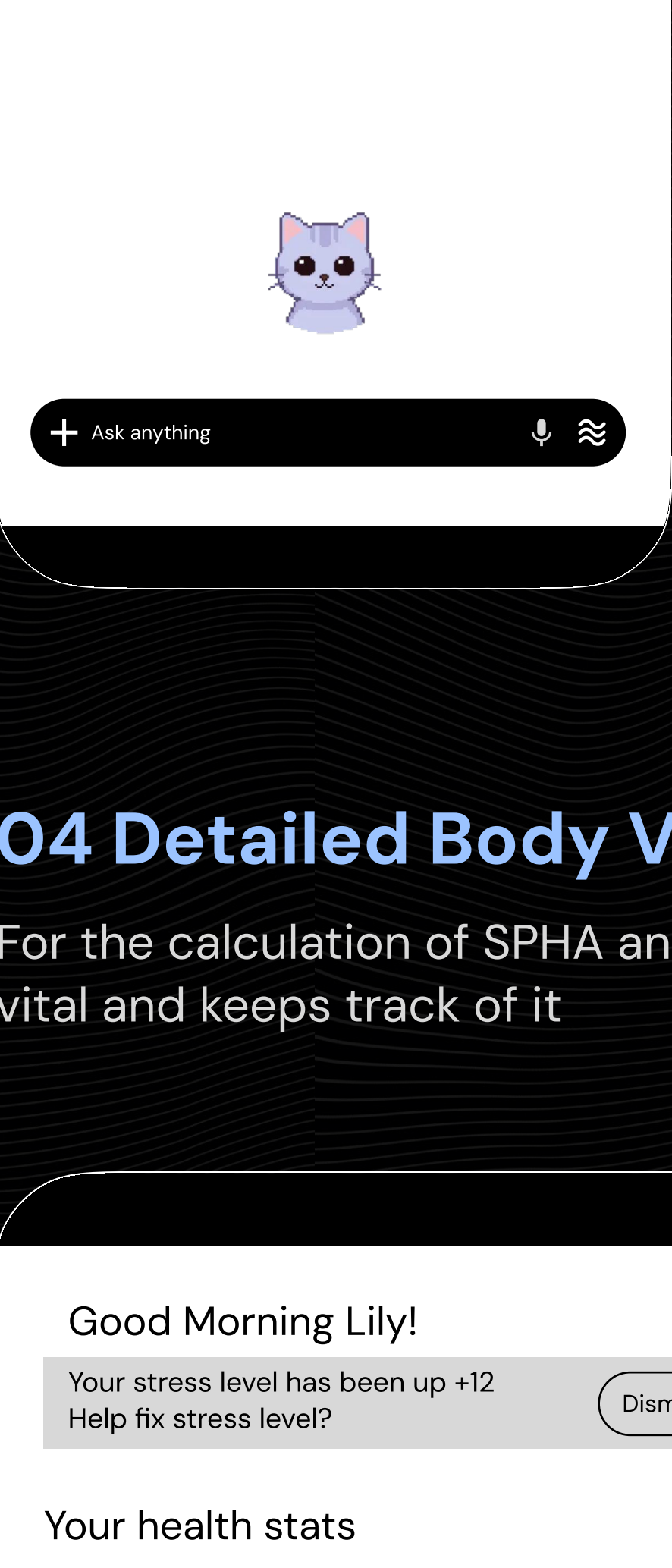
03 Adjust Your Cravings

Instead of feeling guilty on giving up to cravings, it adjust your cravings to suit your goals and take care of the remaining goals



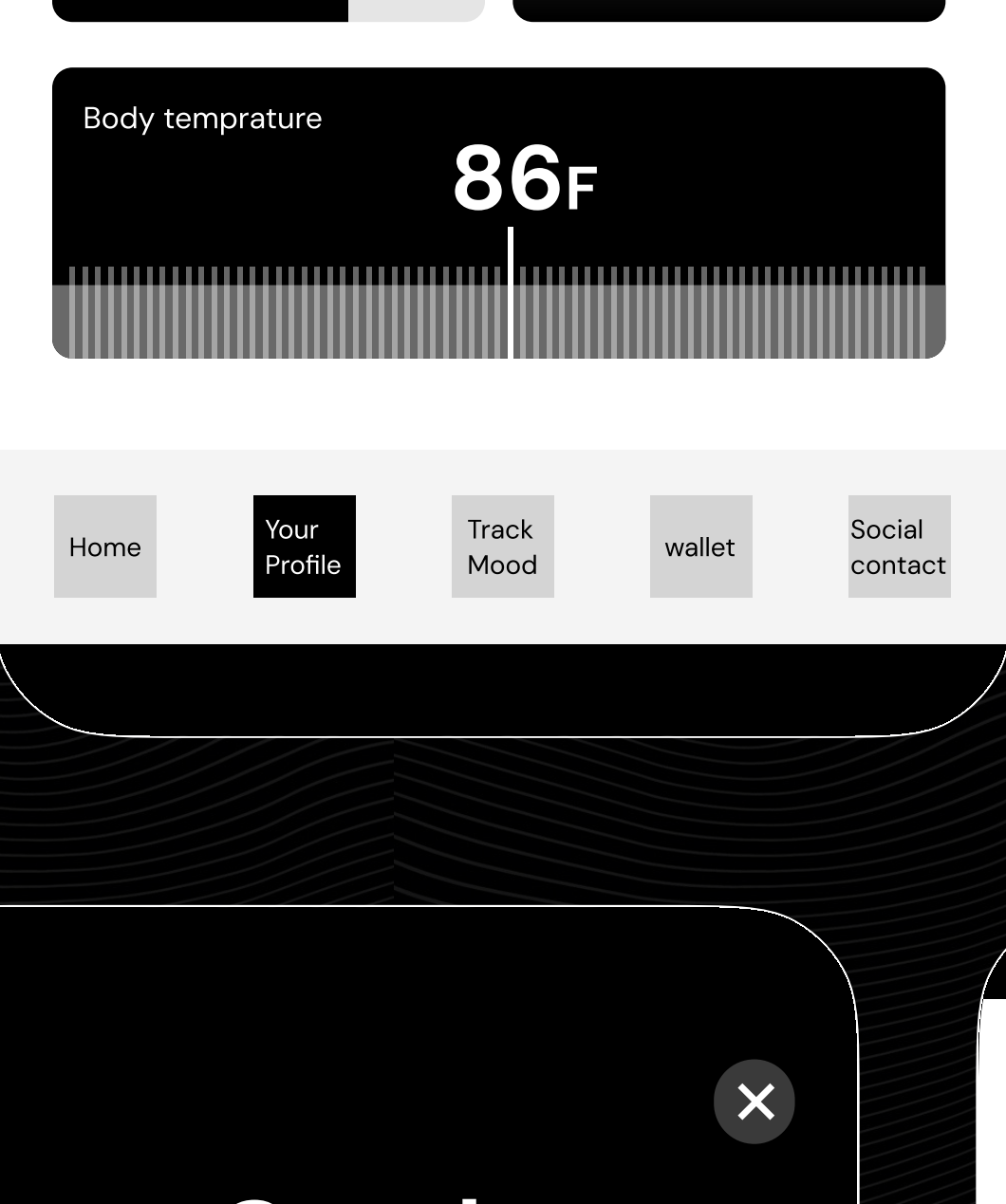
Floating bubble that opens up as a dialogue/ chat window

Data Insight Conversion: People overly rely on AI chat agent ex. chatGPT, Gemini etc for getting personalized food recommendation but ends up not having context of their particular situation

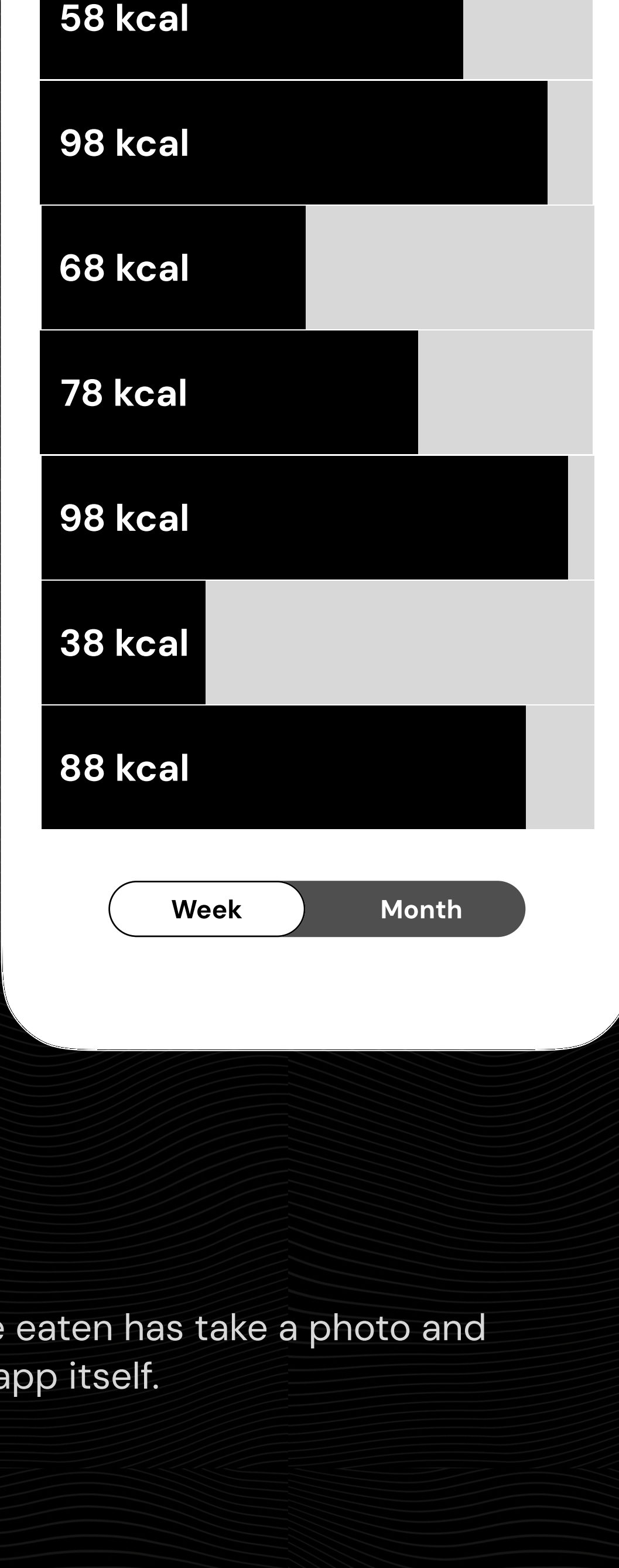
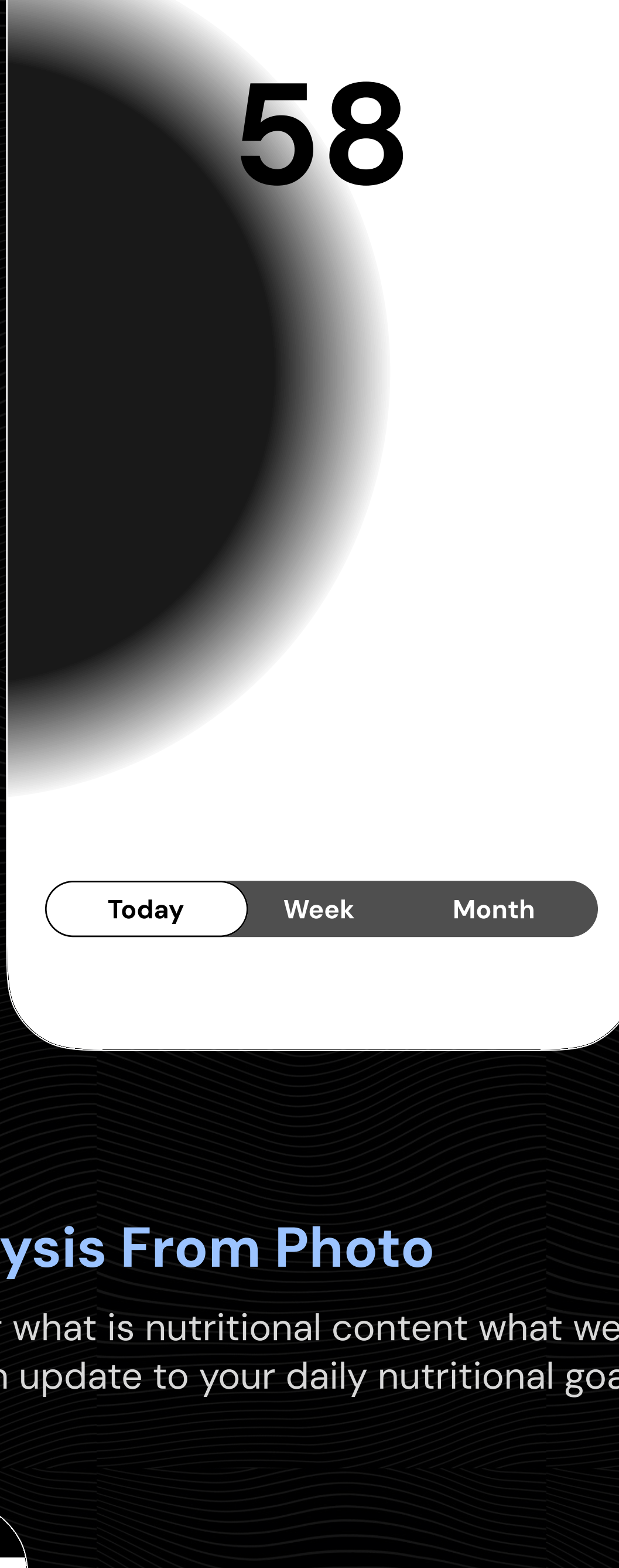
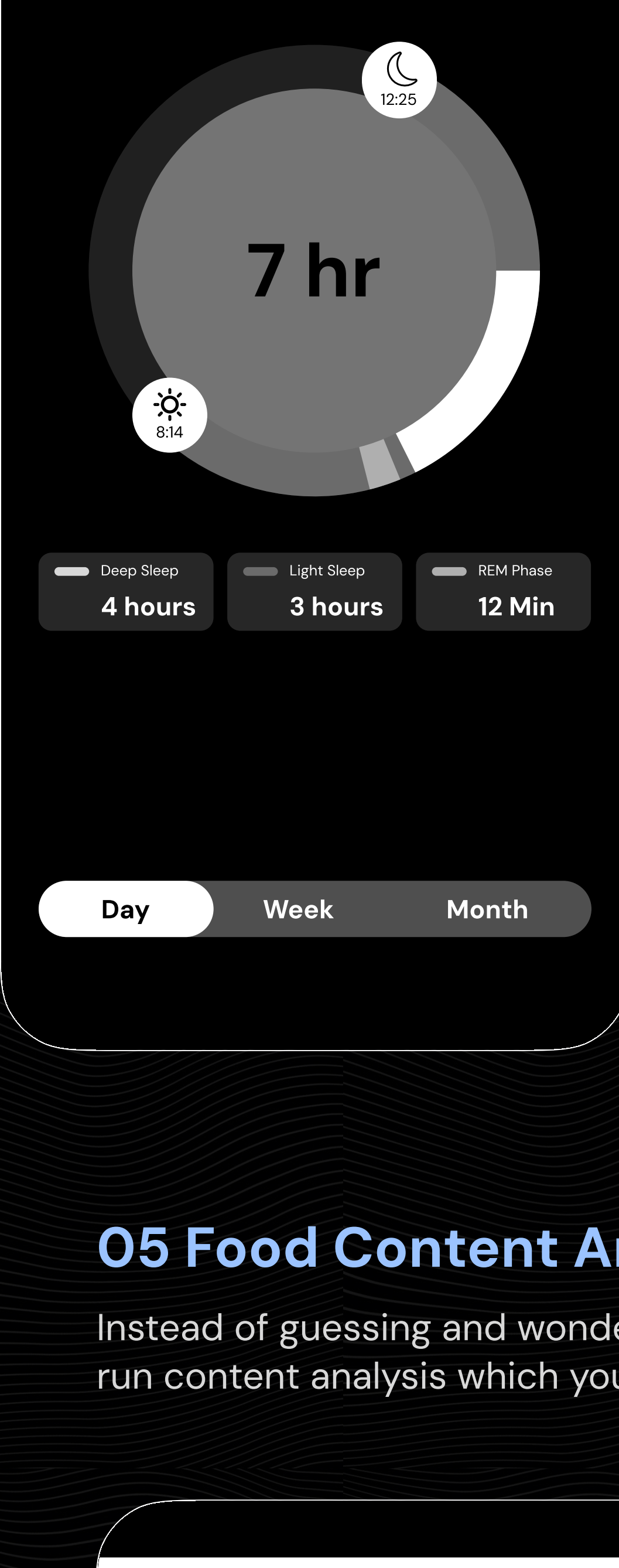


04 Detailed Body Vitals

For the calculation of SPHA and get personalized food suggestion, the app quietly observes the body vital and keeps track of it

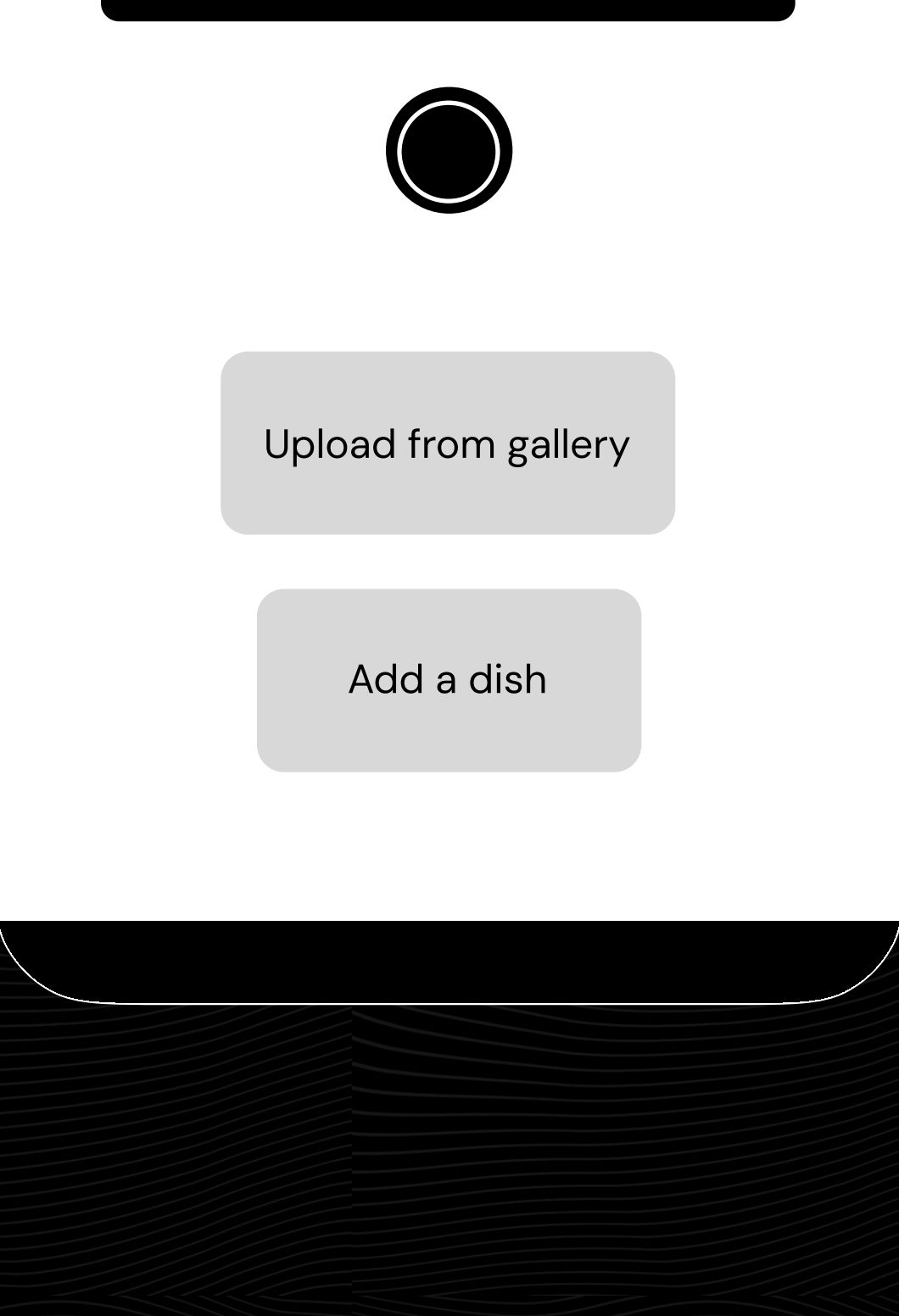


The body vital displayed in a visual manner which reflects their physical nature



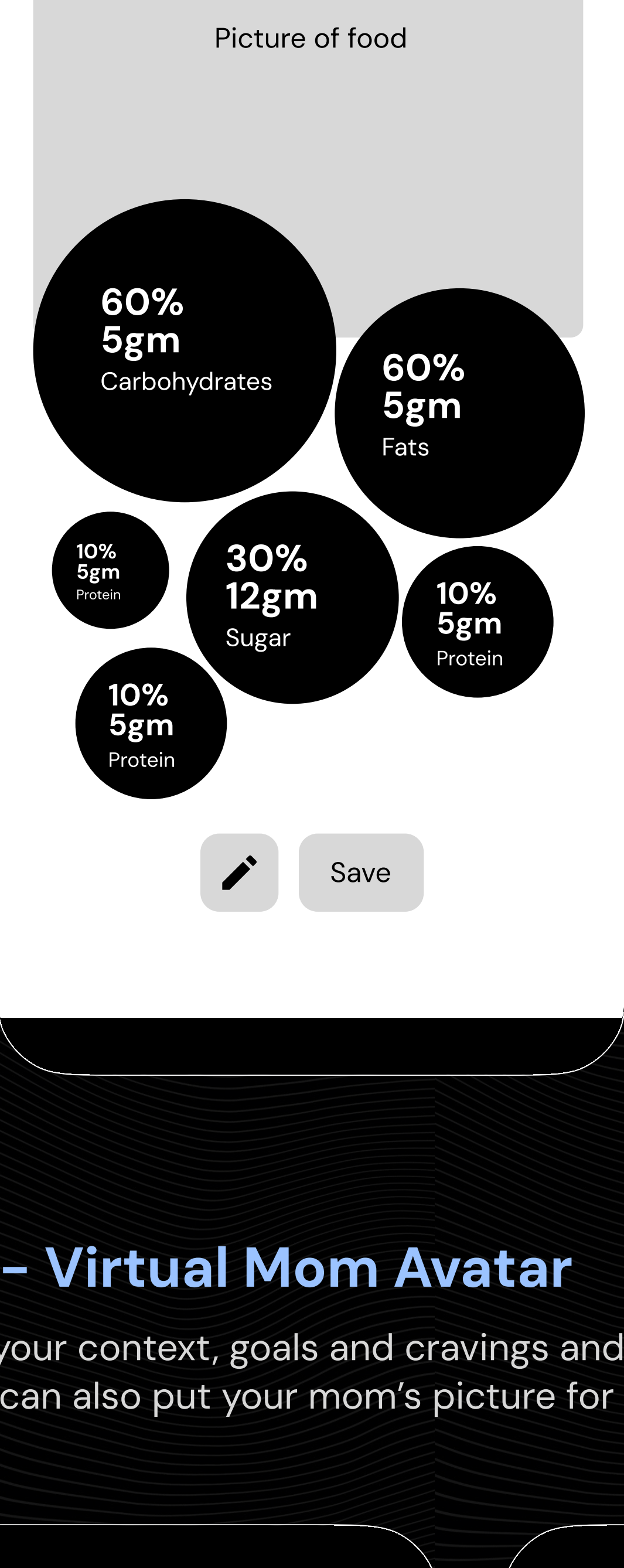
05 Food Content Analysis From Photo

Instead of guessing and wondering what is nutritional content what we have eaten has take a photo and run content analysis which you can update to your daily nutritional goal on app itself.



Data Insight Conversion

Through research we found that people do take photos of their food especially the ones who are on diet to send their consulting dietitians or family group to check up it they are eating correctly

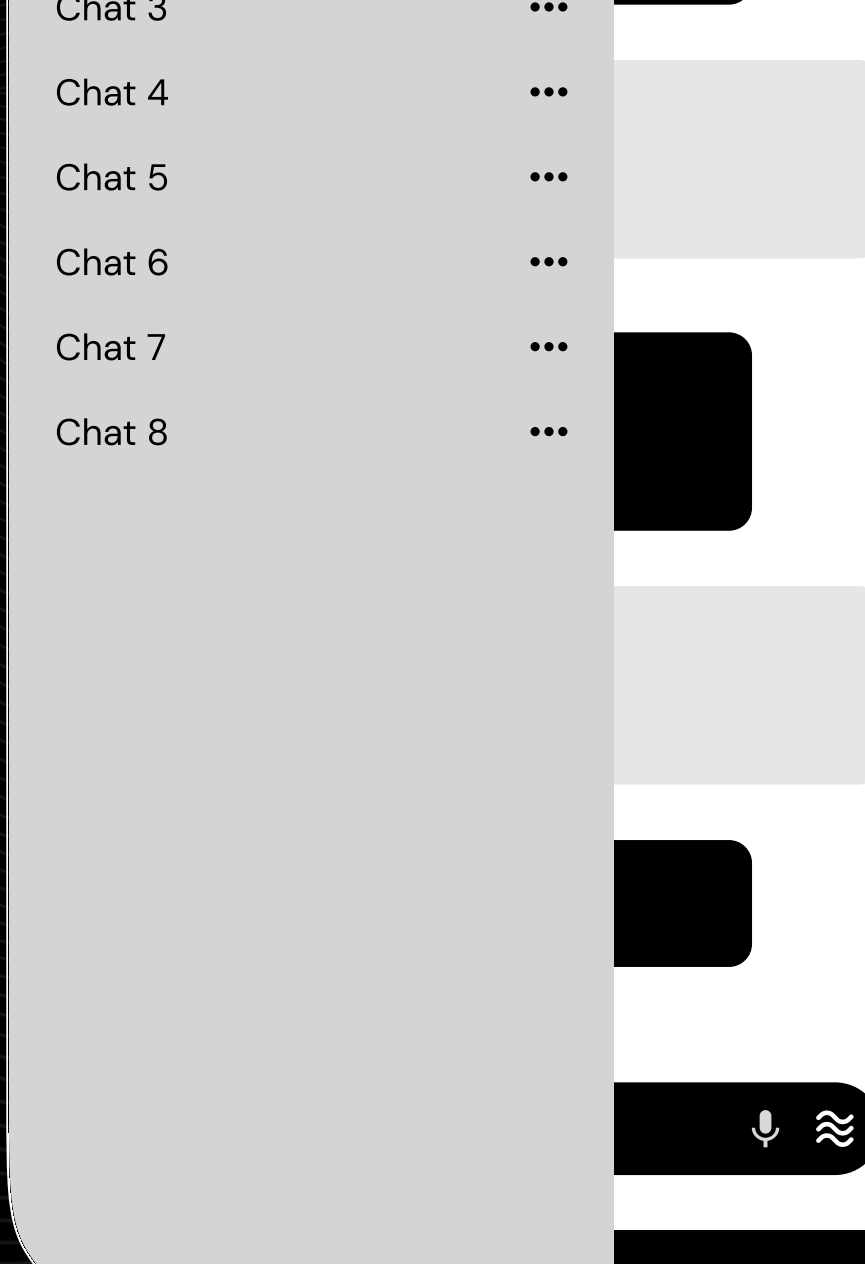
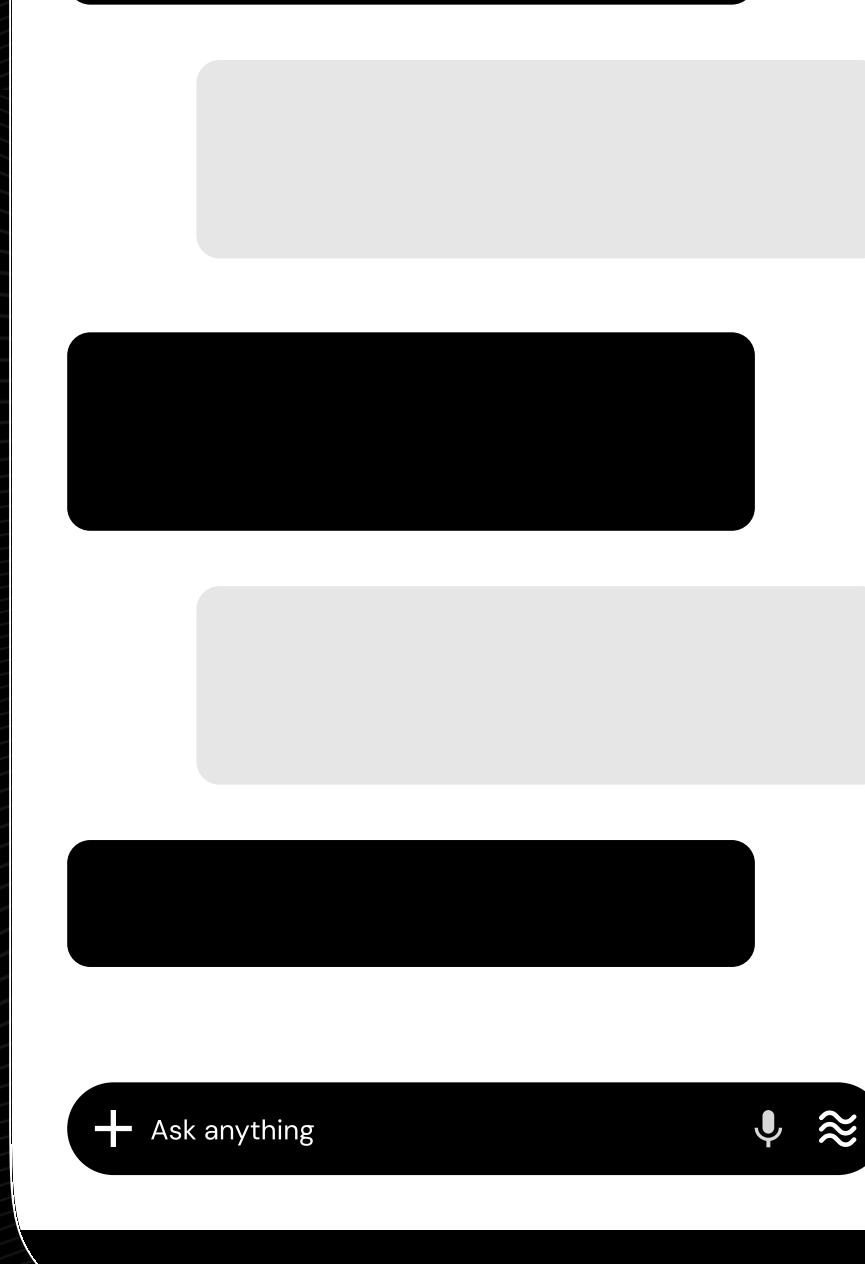
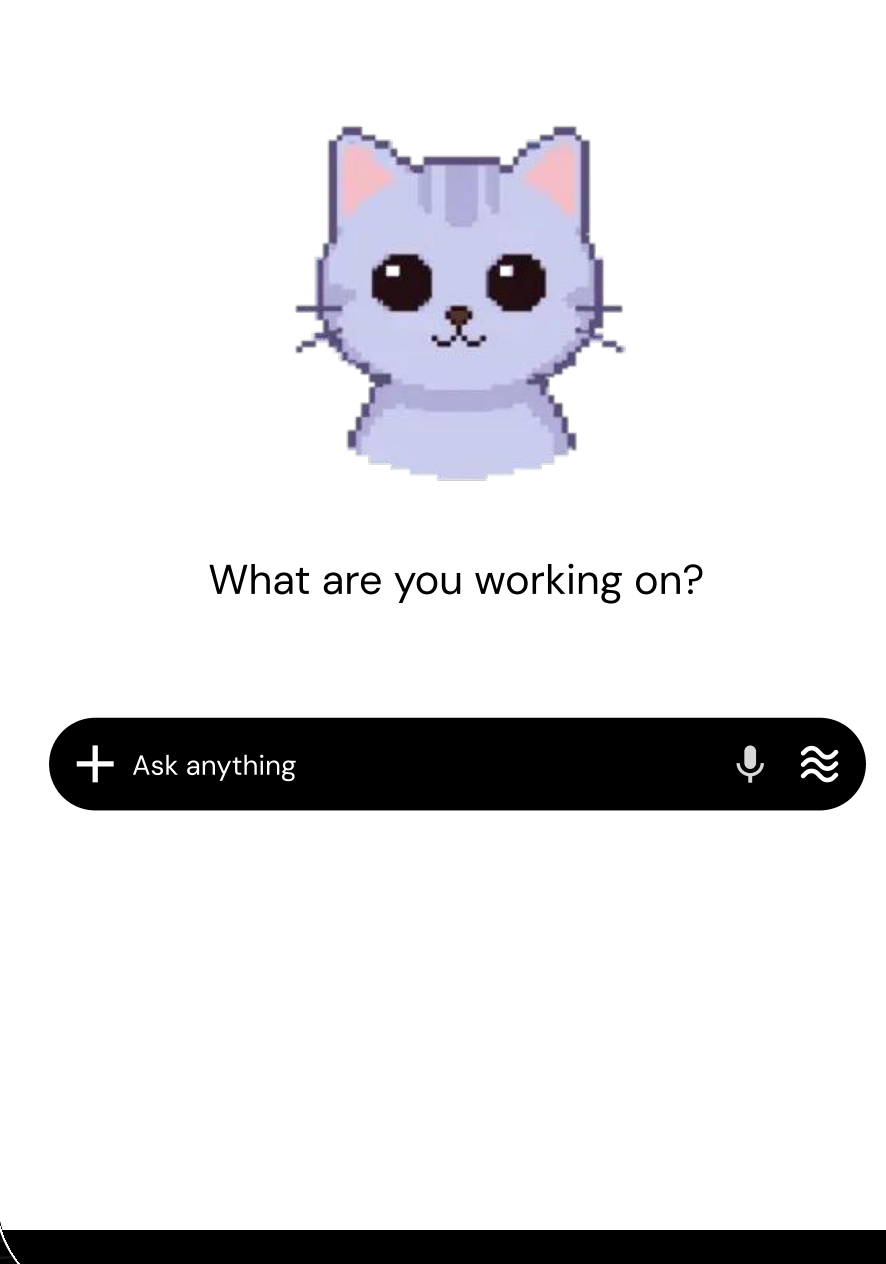
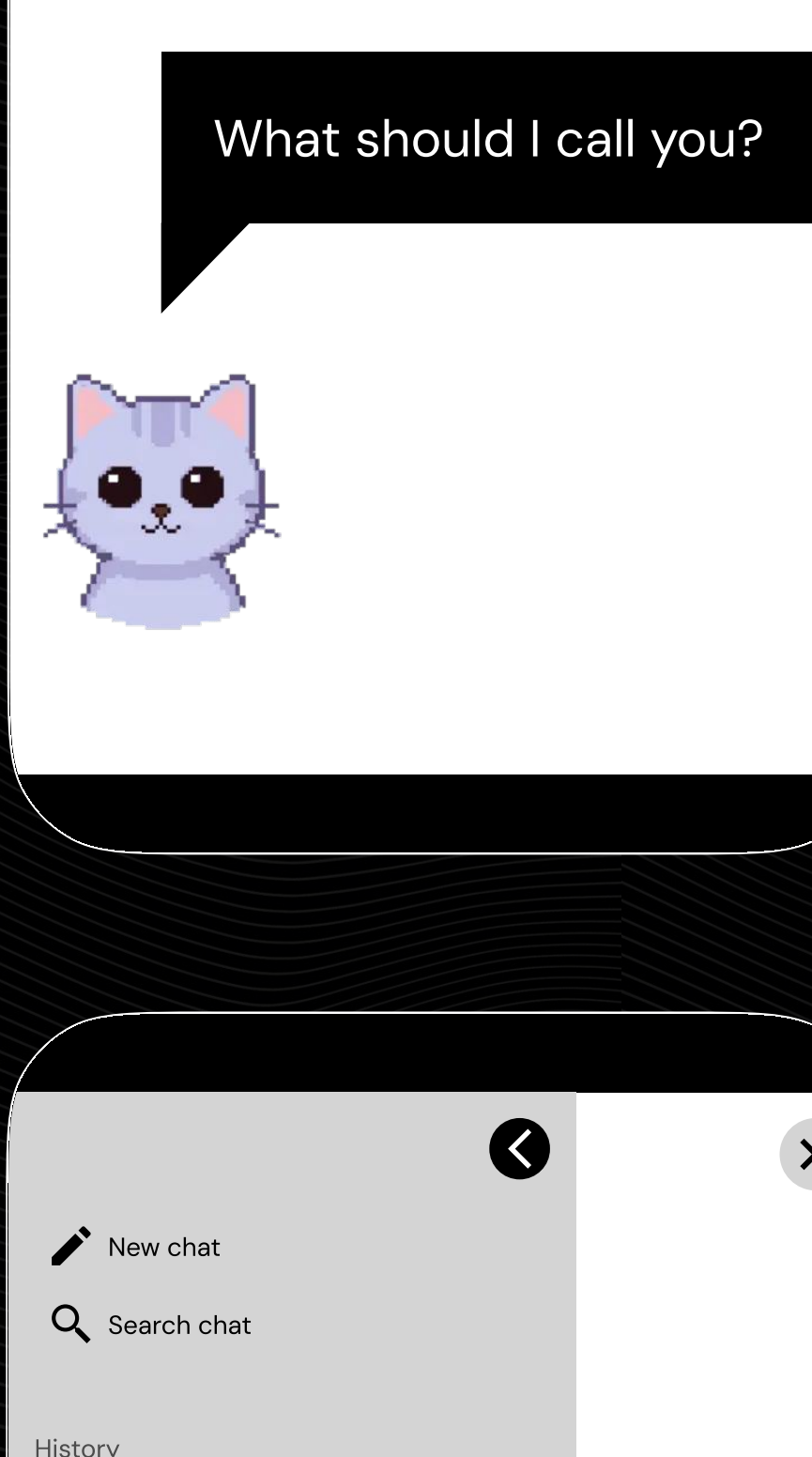
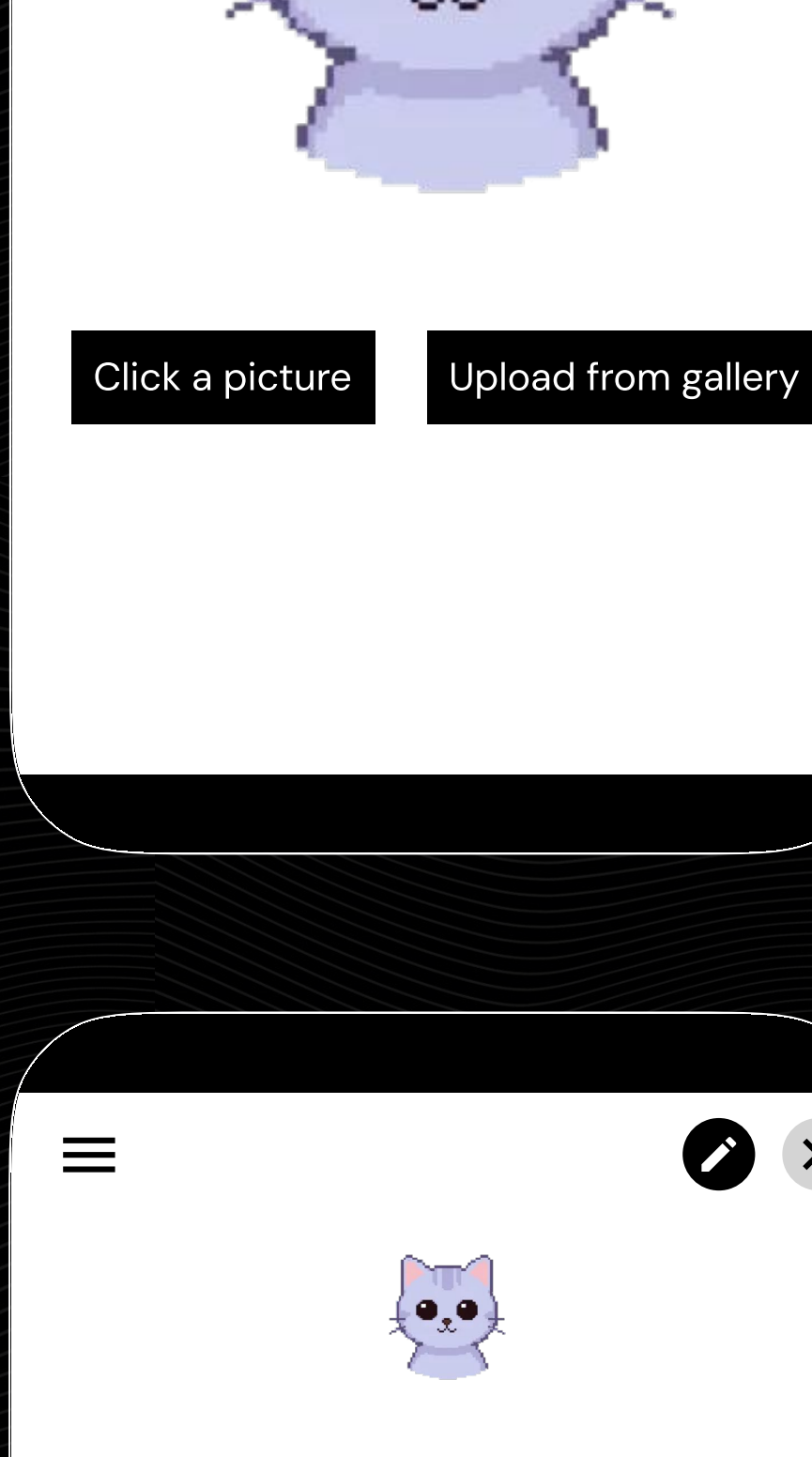
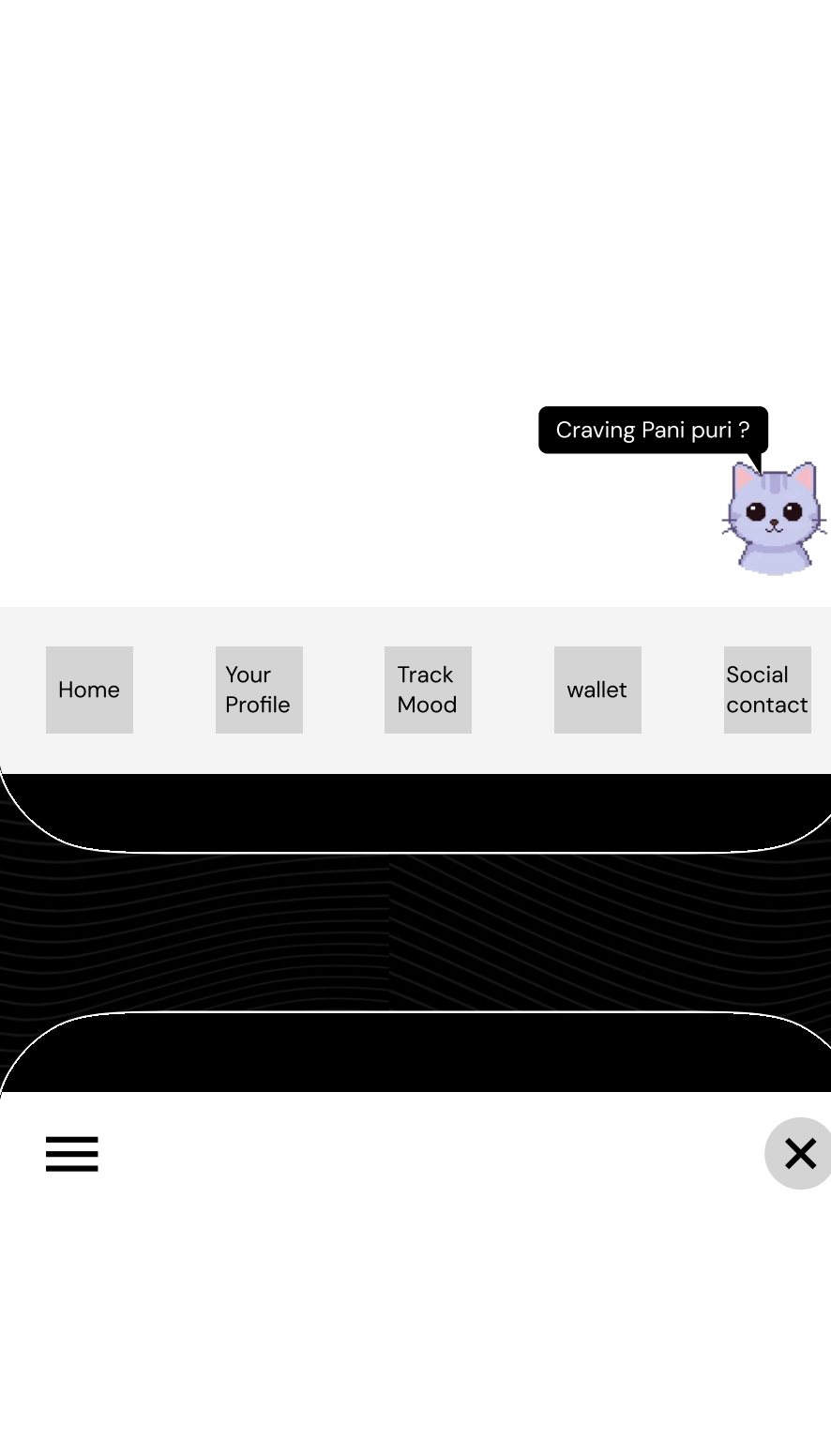


Size reflecting the amount contain in the dish

Sometime we know the details better than machine so to correct that edit option.

06 Live Food Suggestions – Virtual Mom Avatar

Set up an virtual mom that which knows your context, goals and preferences and can talk to related to food and your craving instead of AI chats. You can also put your mom's picture for the avatar!

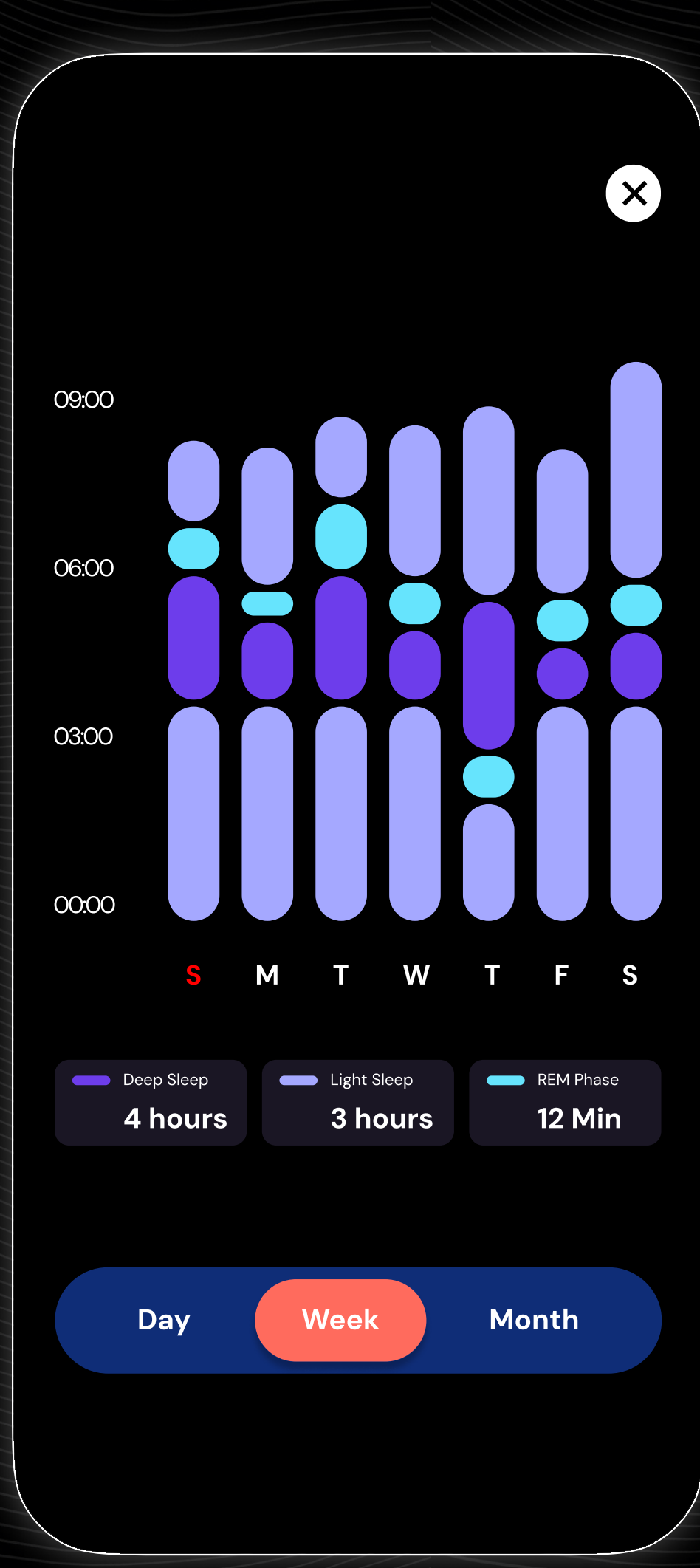
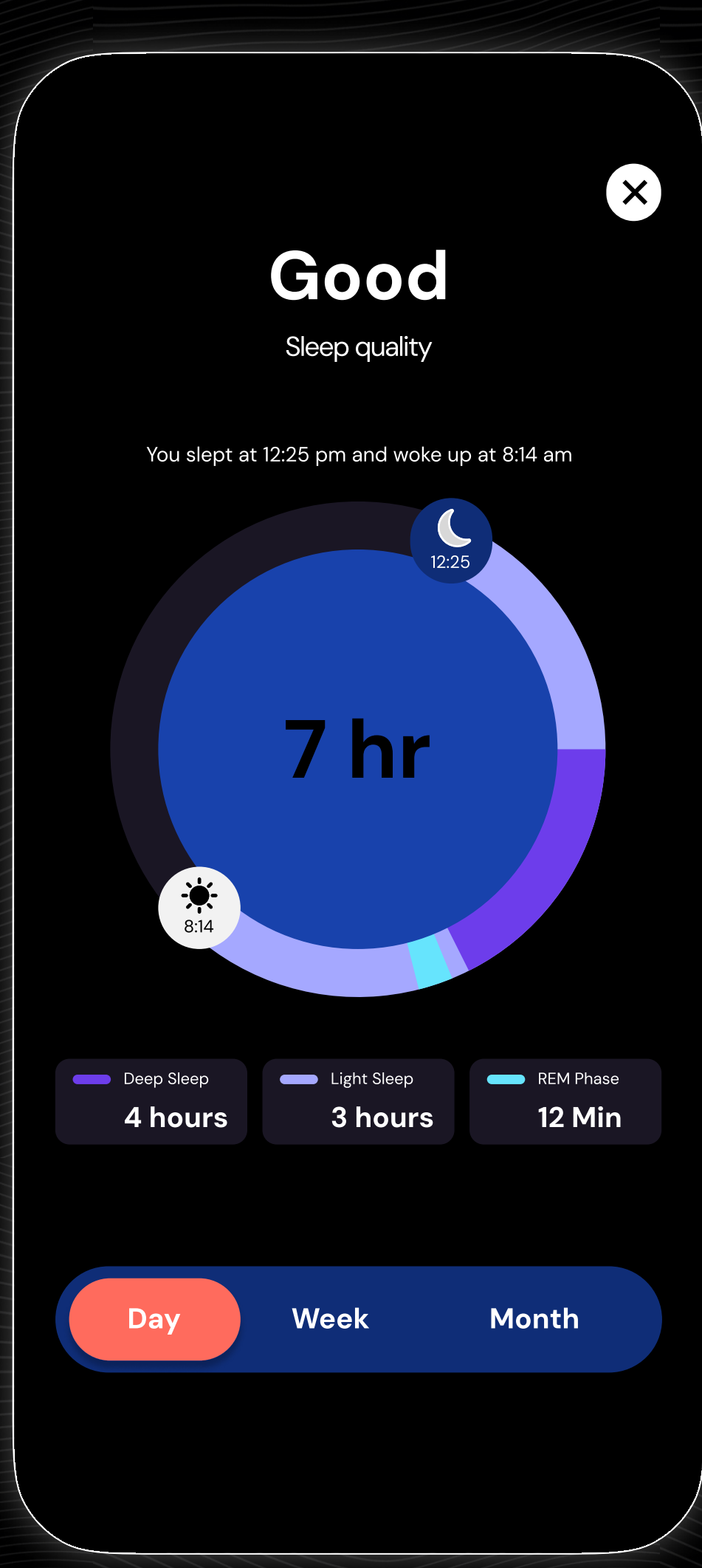
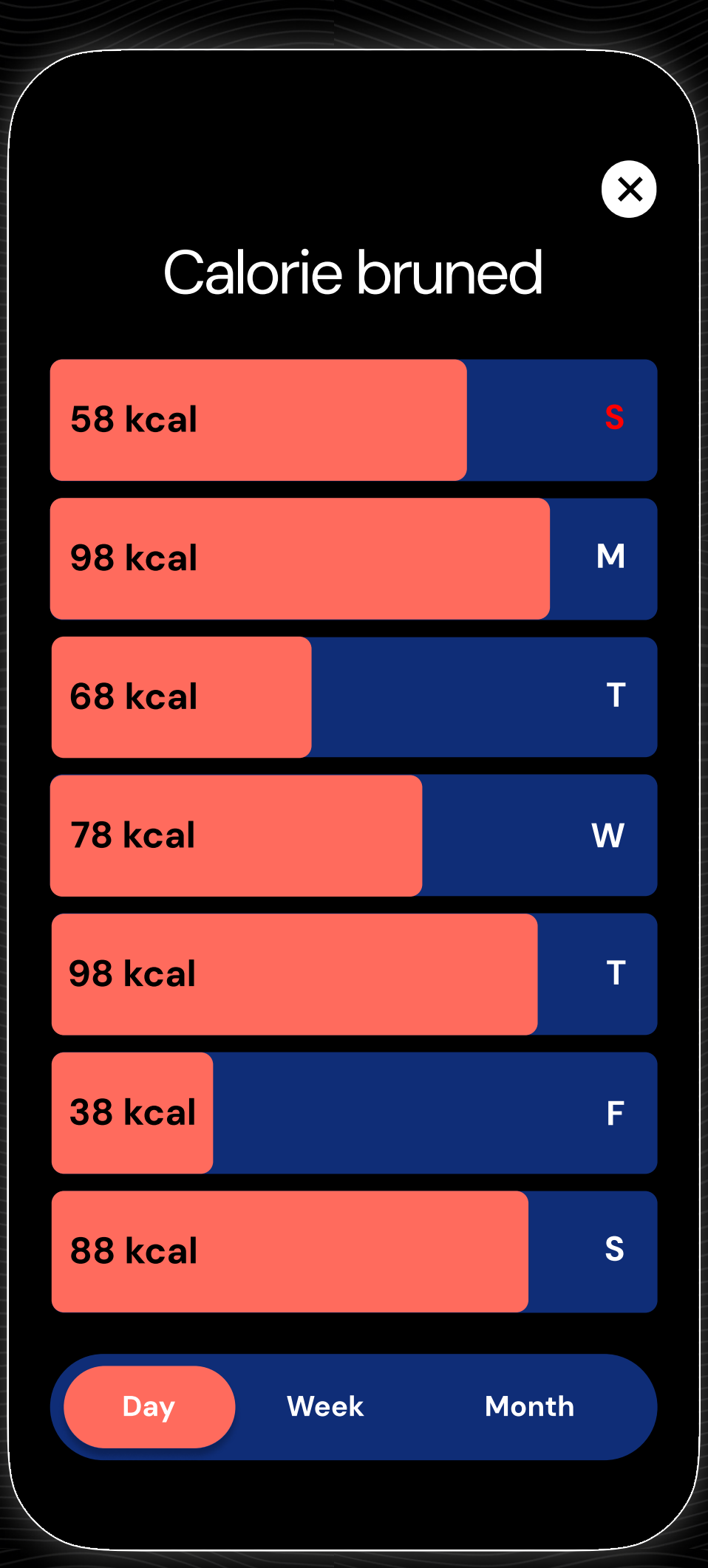
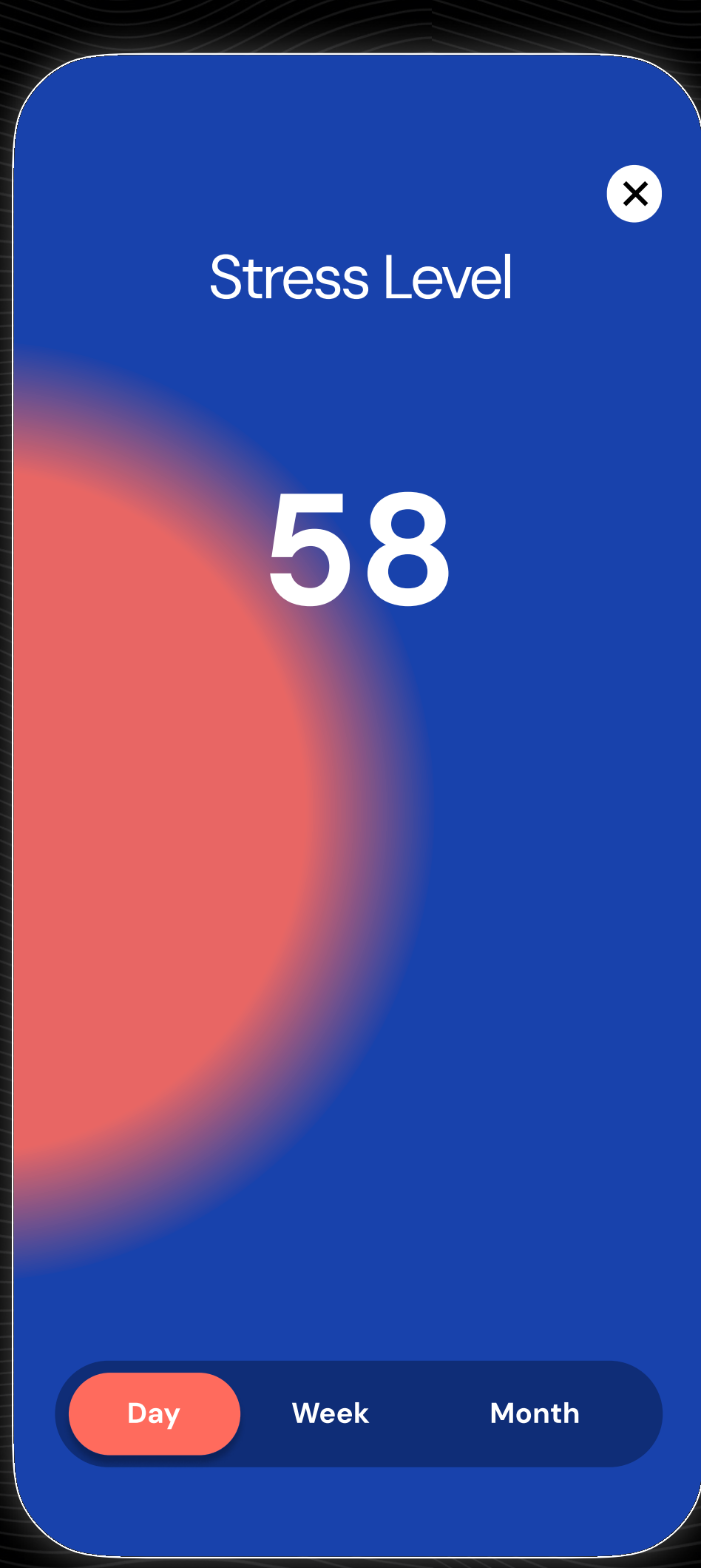
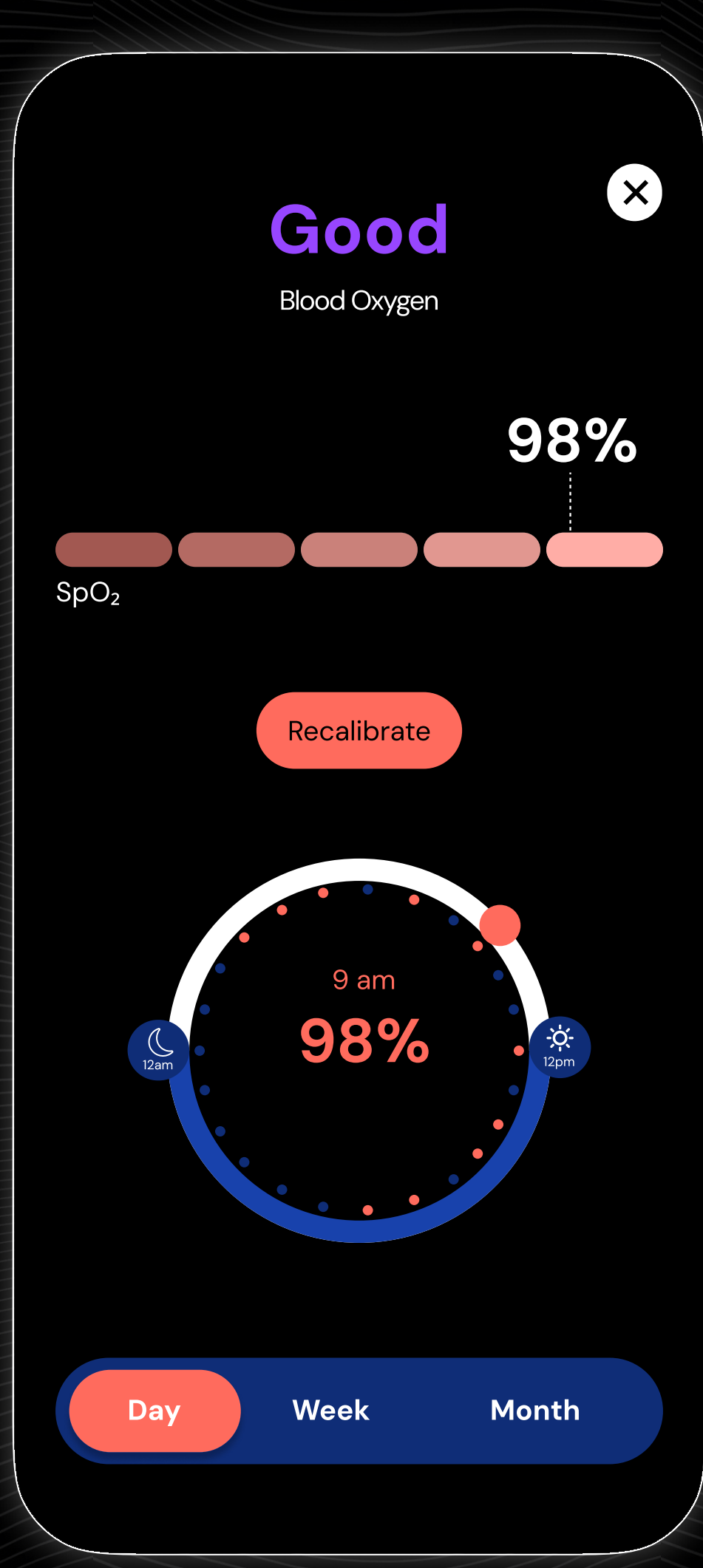
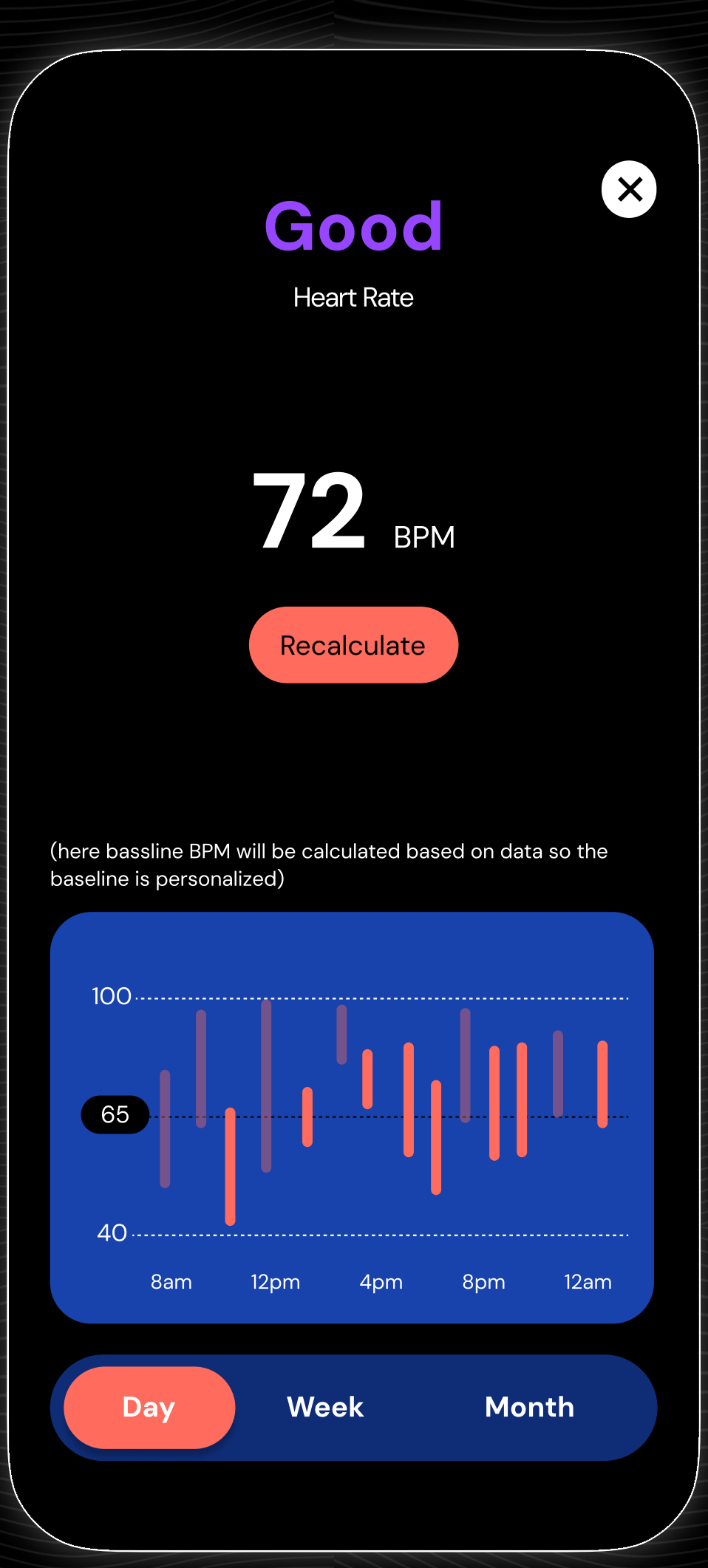
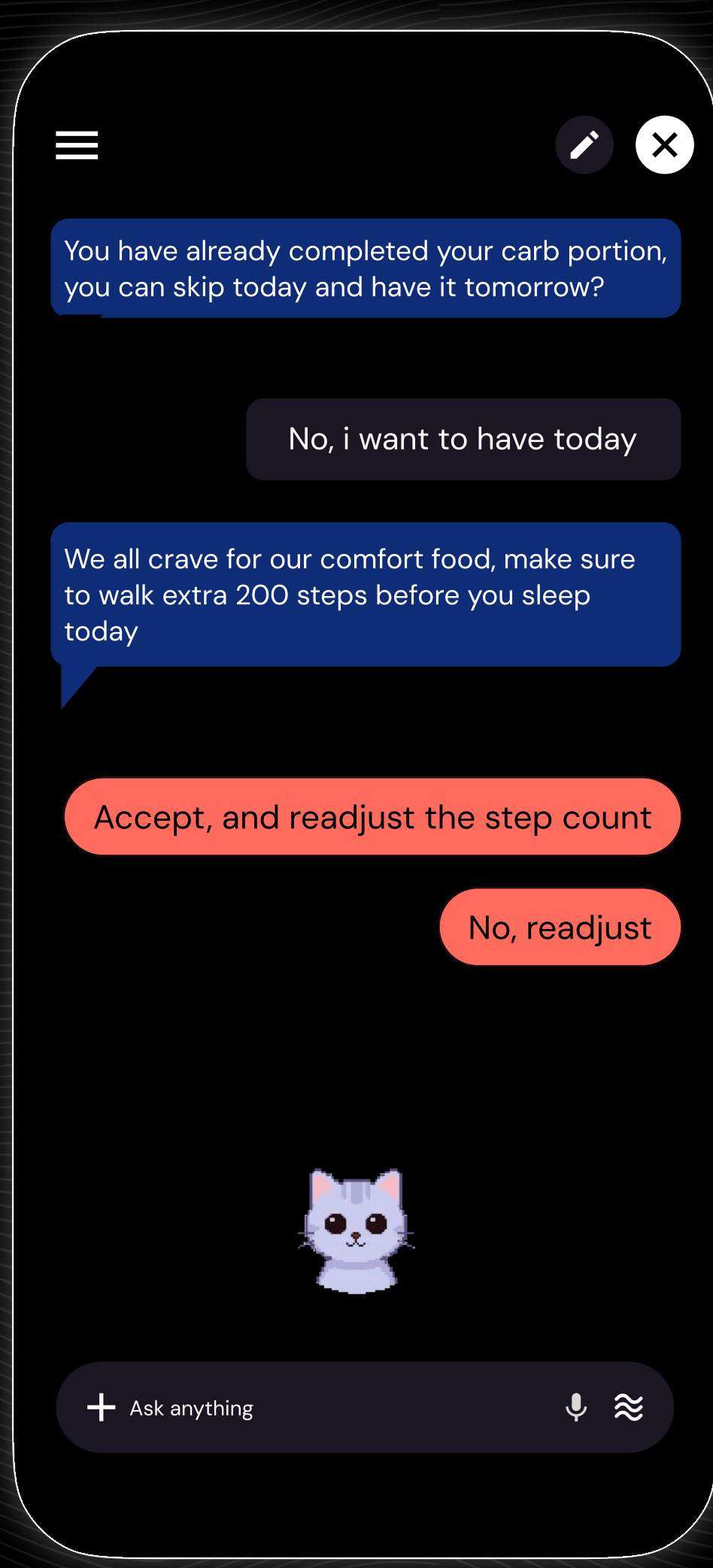
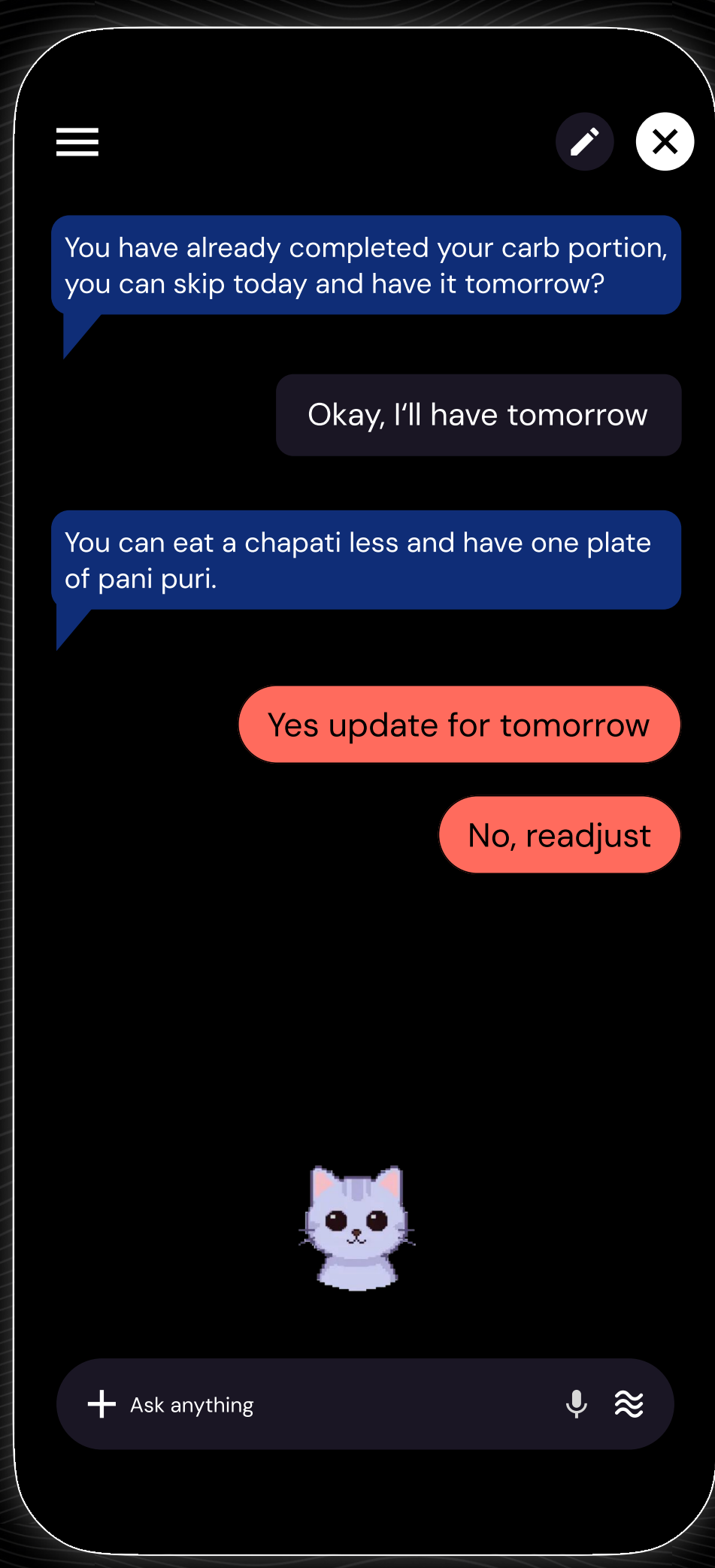
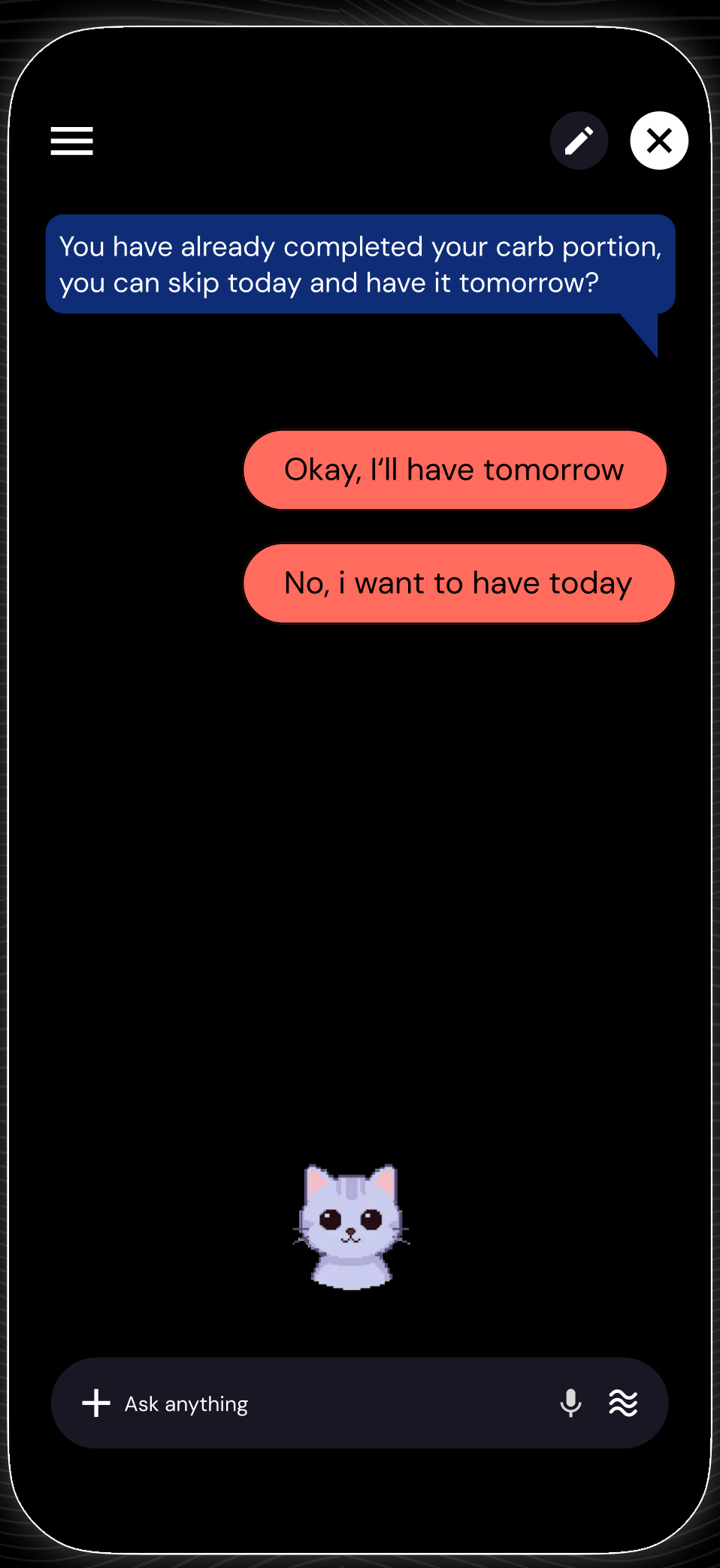
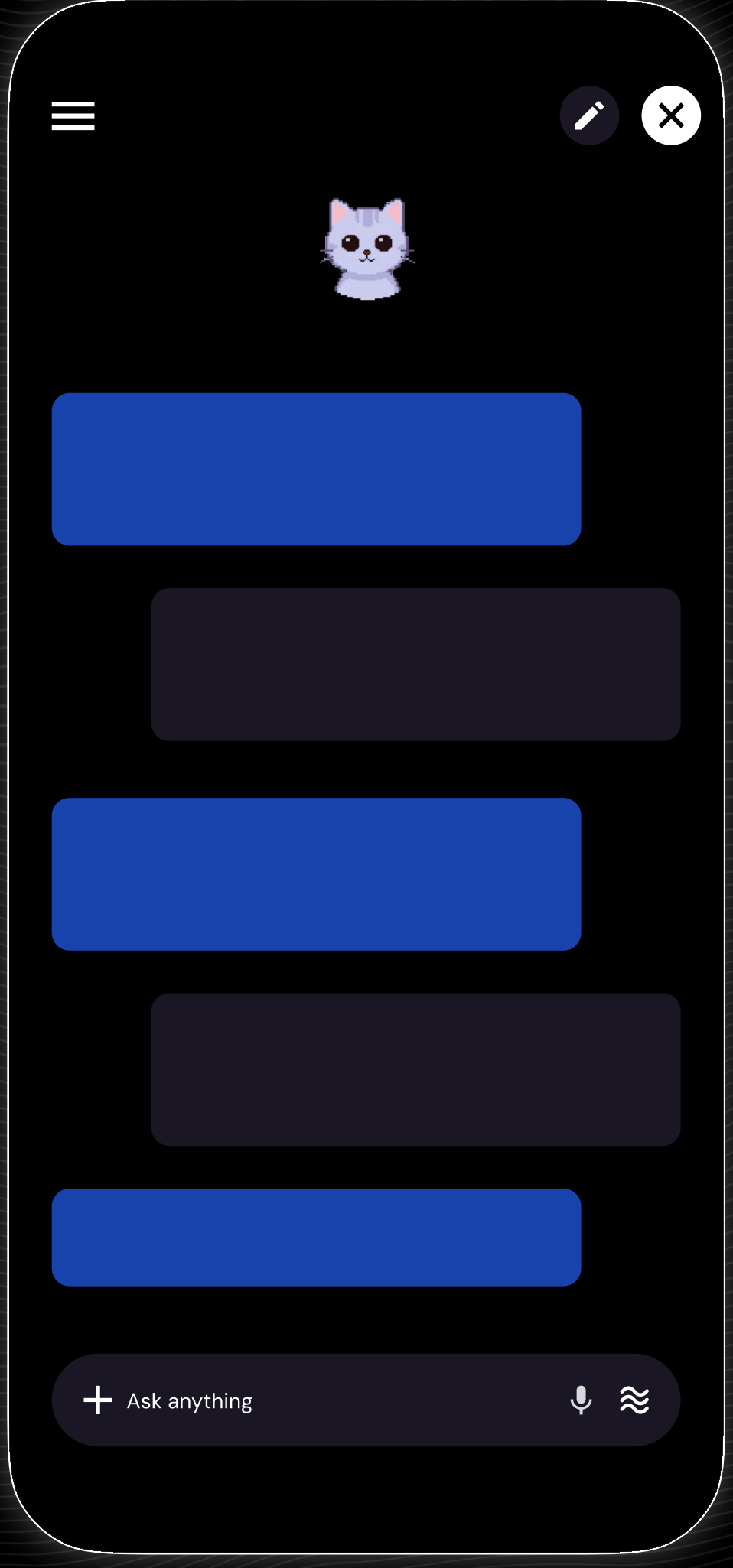
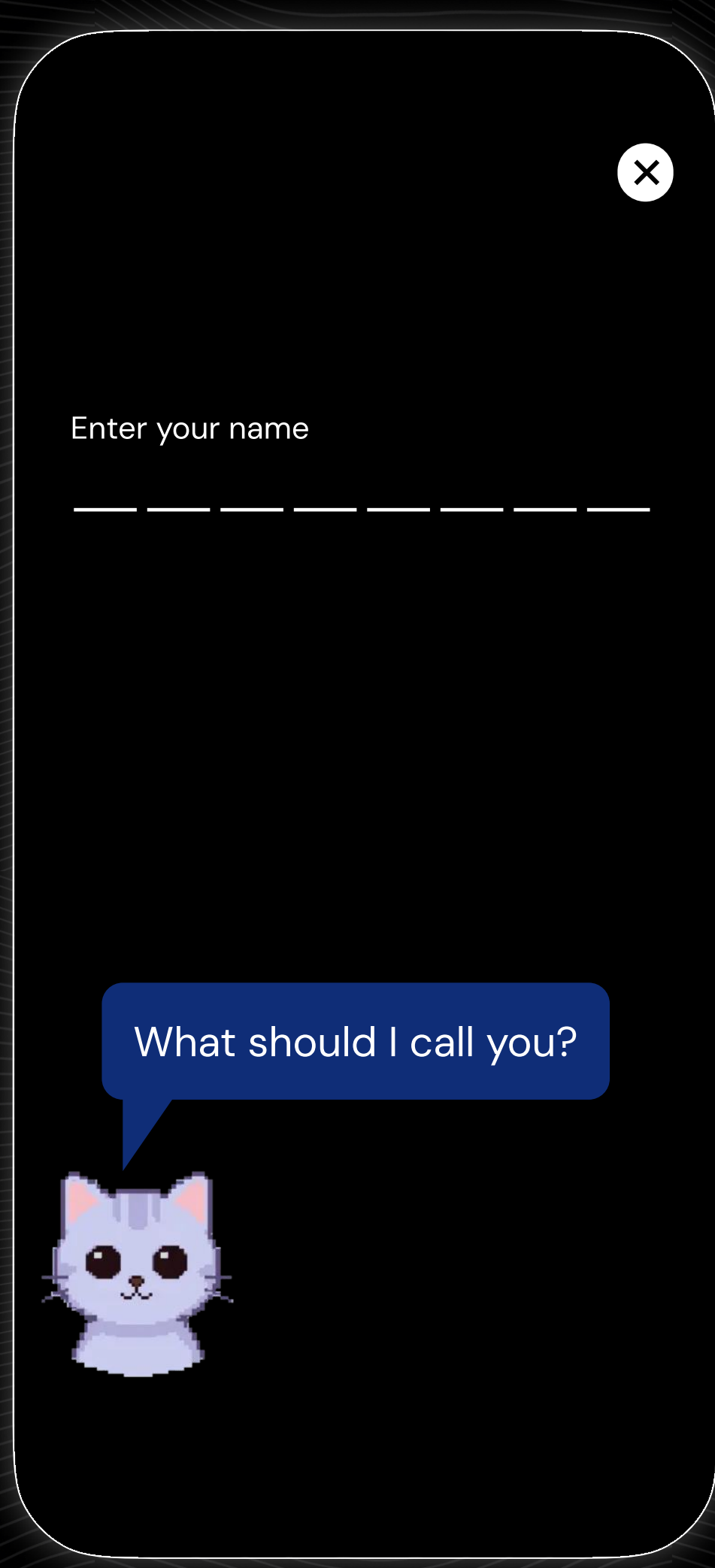
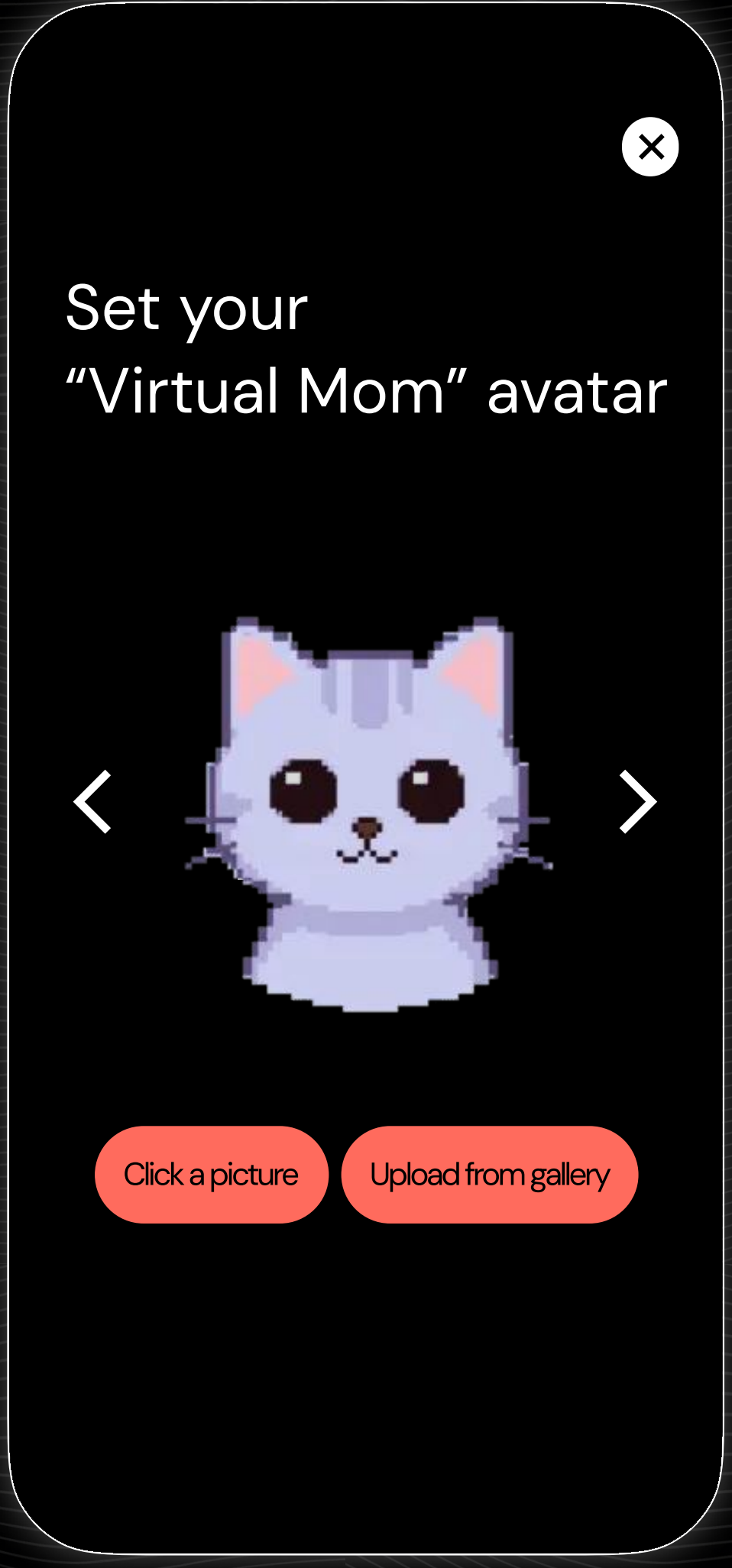
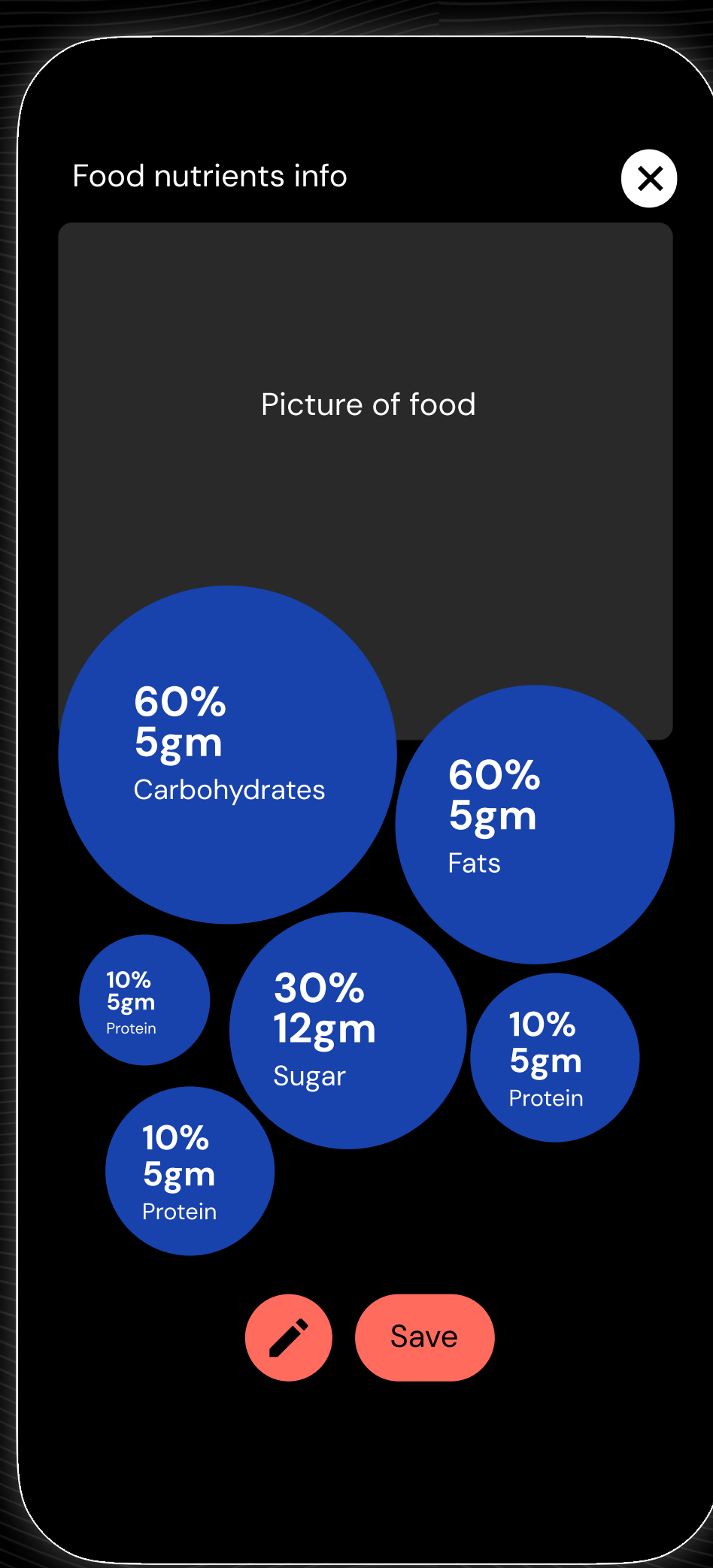
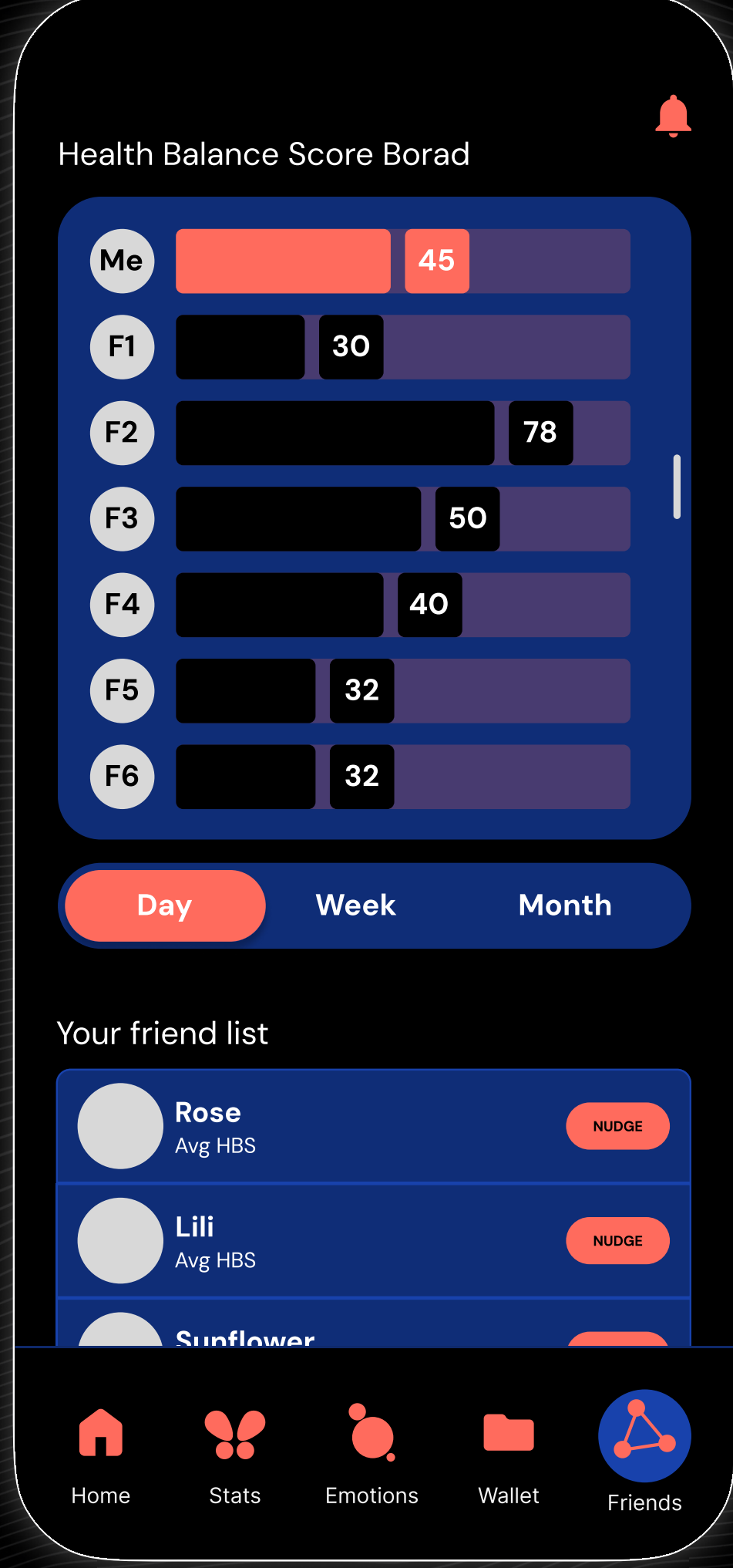
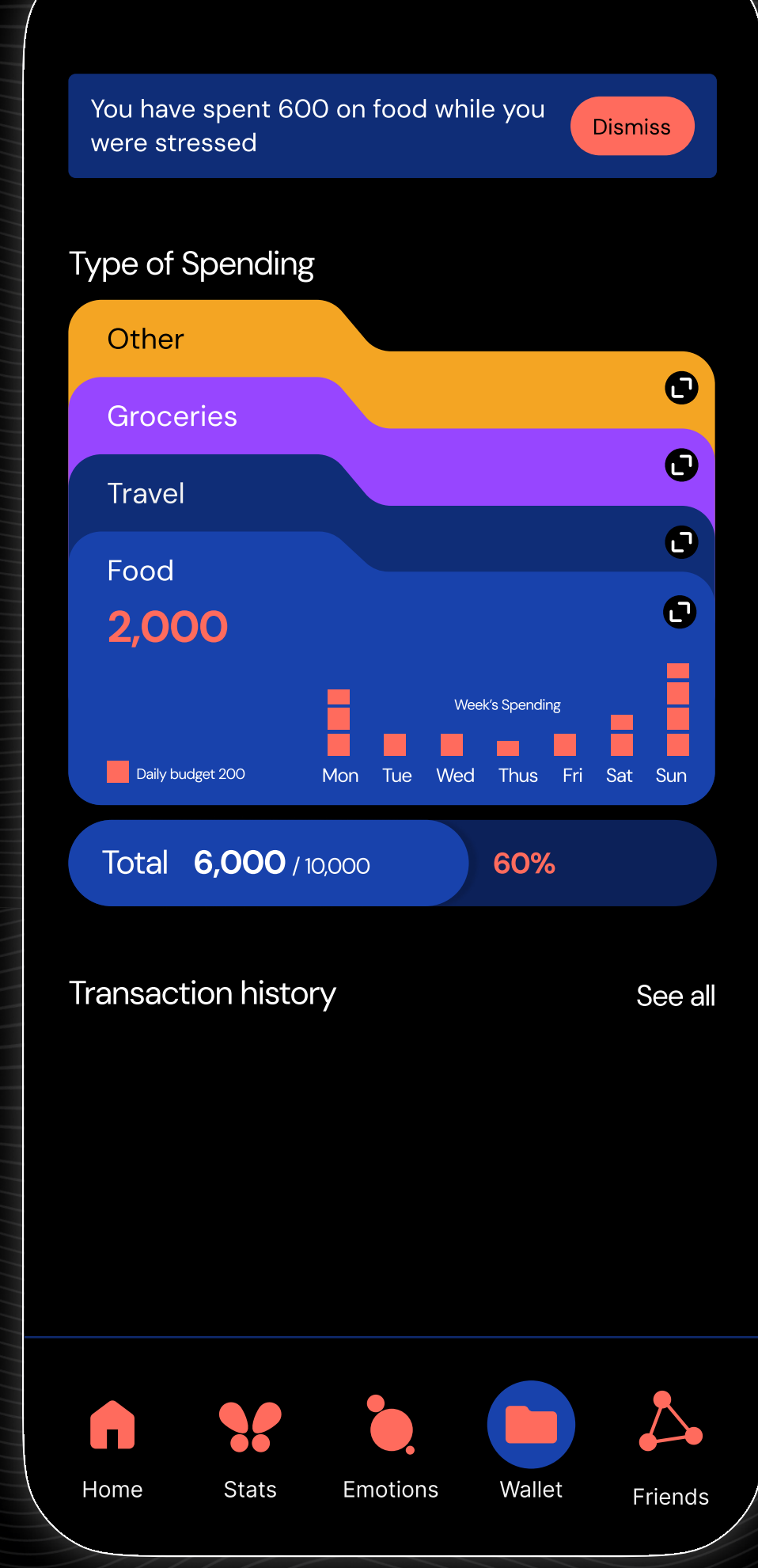
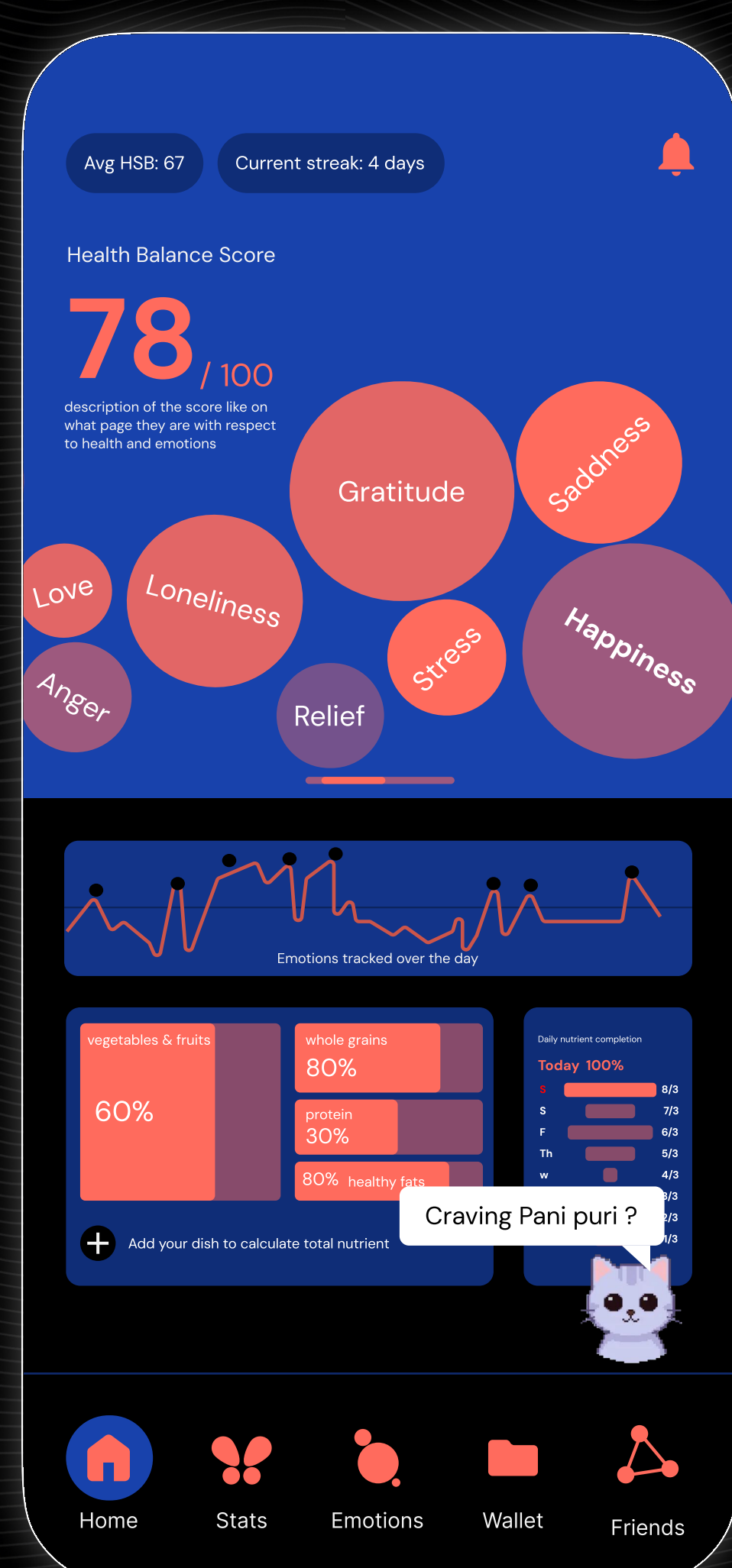


#8 Mood Board

Taking inspiration for y2k UI style an era of taking care and having your own style, translating into UI for this app to install the traditional and caring vibe of that era



#9 UI Screens



#10 Prototype